

# Realized Volatility Forecasting

## A Project Reference

Everything needed to build and defend  
the vol forecasting system

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**Part I**

**The Project**

# Chapter 1

## What We’re Forecasting

Realized variance on day  $t$  is  $RV_t = \sum_{i=1}^M r_{t,i}^2$ , the sum of  $M$  squared intraday returns sampled at 5-minute frequency ([Andersen et al., 2003](#)). Our target is  $\log RV_{t+h}$  for horizons  $h \in \{1, 5, 22\}$  trading days, where multi-day targets aggregate:  $RV_{t+1:t+h} = h^{-1} \sum_{j=1}^h RV_{t+j}$ .

### 1.1 The Universe

The model is trained and evaluated on 35 instruments over 11.3 years of history (Jan 2012 to Mar 2023).

Table 1.1: Forecast universe: 35 instruments.

| Category             | Count     | Examples                        |
|----------------------|-----------|---------------------------------|
| Mega-cap US equities | 30        | AAPL, MSFT, JPM, JNJ, XOM, AMZN |
| Broad-market ETFs    | 4         | SPY, QQQ, IWM, DIA              |
| Equity index futures | 1         | E-mini S&P 500 (ES)             |
| <b>Total</b>         | <b>35</b> |                                 |

All RV series are computed from 5-minute returns using the realized kernel estimator of [Barndorff-Nielsen et al. \(2008\)](#), which corrects for microstructure noise. Models are fit per-asset (no pooling across names).

### 1.2 Success Criteria

The primary metric is QLIKE, a quasi-likelihood loss that penalizes relative forecast errors and is robust to noisy volatility proxies ([Patton, 2011](#)). The formal definition appears in Chapter 13.

### 1.3 The High-Level Pipeline

Figure 1.1 shows the end-to-end forecasting system. Raw data from six sources feeds into a layered feature-engineering pipeline (Layers 0–7, detailed in Chapters 3–8). Two model families consume the resulting feature matrix: LightGBM (Ch. 9) and an LSTM operating on intraday return sequences (Ch. 10). Their forecasts are combined via an ensemble blend (Ch. 11) and evaluated against QLIKE.

The six raw data sources are: (i) tick-level trade prices, (ii) daily OHLCV, (iii) E-mini S&P 500 Level 2 order book, (iv) SPX implied volatility surface, (v) cross-asset prices (VIX, bonds, gold, oil, FX), and (vi) earnings and macro event calendars.

Table 1.2: Evaluation criteria.

| Priority  | Metric               | Requirement                                                                                        |
|-----------|----------------------|----------------------------------------------------------------------------------------------------|
| Primary   | QLIKE                | 30–80 bps improvement over HARQ baseline (Bollerslev et al., 2016), averaged across universe       |
| Secondary | MSE, MAE             | Reported alongside QLIKE for robustness                                                            |
| Secondary | Diebold-Mariano test | Statistically significant improvement ( $p < 0.05$ ) vs. each baseline (Diebold and Mariano, 1995) |
| Tertiary  | Economic value       | Out-of-sample utility gain in a volatility-targeting portfolio (Moreira and Muir, 2017)            |

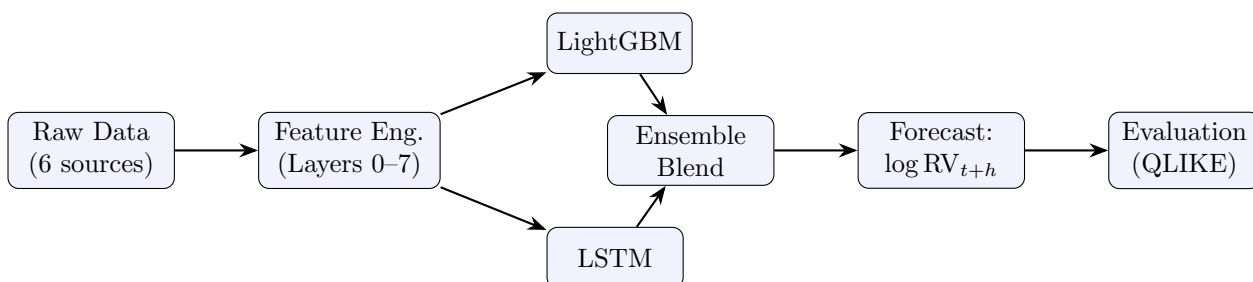


Figure 1.1: High-level forecasting pipeline.

### Features Over Models

The feature set matters more than the model. Layers 0–2 (approximately 20 features covering HAR components, jump decomposition, and signed volatility) achieve 85% of the forecasting accuracy attainable with the full 120-feature set.

## Chapter 2

# Our Data

### 2.1 Data Sources

Table 2.1 inventories every data source available for the project. All sources span 11.3 years of history.

Table 2.1: Available data sources, what they enable, and their constraints.

| Data Source                               | Granularity                     | What It Enables                               | Key Constraint                                                |
|-------------------------------------------|---------------------------------|-----------------------------------------------|---------------------------------------------------------------|
| Tick-level RV (34 symbols)                | L1 tick                         | RV at any frequency, RQ, jumps, semivariances | —                                                             |
| Daily OHLCV (34 + VIX)                    | Daily                           | HAR baselines, ML training                    | No intraday structure                                         |
| E-mini L2 depth                           | L2 (~4M ticks/day)              | OBI, depth ratio, VPIN, LSTM input            | E-mini only                                                   |
| IV surfaces (Marquee EDRVOL)              | Full tenor $\times$ strike grid | VRP, skew, term structure, butterfly          | Full surface SPX only; single-stock ATM IV via EDRVOL_PERCENT |
| VIX term structure                        | Daily                           | Regime detection, contango/backwardation      | —                                                             |
| Cross-asset (treasuries, FX, commodities) | Mixed (tick to daily)           | Spillover features, macro regime              | Daily for some instruments                                    |

### 2.2 GS Edge vs. Public Data

Table 2.2 summarizes the capabilities our data provides relative to what is typically available in academic studies.

#### The data advantage is structural, not just bigger

The tick-level RV, full IV surface, and L2 depth data let us construct features that are mathematically impossible to compute from public data. The gap is one of kind rather than volume: no amount of additional public data would allow the feature set to be replicated.

### 2.3 Constraints That Shape Decisions

L2 depth data covers the E-mini only, so microstructure depth features (OBI at levels 2–5, depth ratio) apply exclusively to the index. Equities receive L1 features only. The full IV surface is SPX only, so surface-derived features (skew, term structure, butterfly) function as market-wide regime signals. Single-stock ATM IV is available via Marquee EDRVOL\_PERCENT, enabling per-name VRP and



Table 2.2: Data advantages over standard academic sources.

| Capability            | Our Data                                      | Public Alternative                  | Edge                                                            |
|-----------------------|-----------------------------------------------|-------------------------------------|-----------------------------------------------------------------|
| RV estimation quality | Tick-level, 34 symbols, 11.3 years            | 5-min returns from TAQ / Oxford-Man | Precise RQ, jump detection, kernel estimators                   |
| Options surface       | Full SPX tenor $\times$ strike grid (Marquee) | VIX only (model-free 30-day)        | Full surface derivatives: skew, butterfly, slope, event-implied |
| Microstructure depth  | E-mini L2 (4M ticks/day)                      | L1 quotes only (TAQ)                | OBI at depth levels 2–5, true depth imbalance                   |
| Cross-asset sync      | Same tick timestamp across asset classes      | Daily closes only                   | Intraday lead-lag detection                                     |
| Panel breadth         | 30 mega-caps + 4 ETFs + E-mini                | Typically 1 index or 29 DJIA        | Graph models, sector structure, cross-sectional features        |

IV–RV gap features. These constraints determine which features apply to which assets in the feature engineering pipeline.

#### Do not treat depth or surface features as stock-level predictors

L2 depth and the full IV surface are index-level data. Using them as if they vary across individual equities would introduce look-ahead bias or a constant signal masquerading as stock-specific information. Single-stock ATM IV is the exception: it is genuinely per-name.

## Part II

# The Feature Set

## Chapter 3

# HAR Core and Measurement Quality

The HAR baseline and its noise-aware extension HARQ form Layer 0 of the feature pipeline. Five features, three horizons, one interaction term.

### 3.1 The Five Foundation Features

Table 3.1: Layer 0 feature set: the five foundation features.

| Feature                                      | Definition                                                      | Role in Forecast                                                        |
|----------------------------------------------|-----------------------------------------------------------------|-------------------------------------------------------------------------|
| $\log \text{RV}_t^{(d)}$                     | $\log \text{RV}_t$                                              | Strongest single predictor; log transform gaussianizes the distribution |
| $\log \text{RV}_t^{(w)}$                     | $\log\left(\frac{1}{5} \sum_{i=0}^4 \text{RV}_{t-i}\right)$     | Medium-memory component; smooths daily noise                            |
| $\log \text{RV}_t^{(m)}$                     | $\log\left(\frac{1}{22} \sum_{i=0}^{21} \text{RV}_{t-i}\right)$ | Long-memory regime anchor                                               |
| $\text{RQ}_t$                                | $\frac{M}{3} \sum_{i=1}^M r_{t,i}^4$                            | Measures how noisy today's RV estimate is                               |
| $\sqrt{\text{RQ}_t} \cdot \text{RV}_t^{(d)}$ | RQ interaction                                                  | Shrinks daily weight on noisy days; highest-impact HAR extension        |

The baseline HAR model (Corsi, 2009) regresses future log-RV on three horizons:

$$\log \text{RV}_{t+1} = \beta_0 + \beta_d \log \text{RV}_t^{(d)} + \beta_w \log \text{RV}_t^{(w)} + \beta_m \log \text{RV}_t^{(m)} + \varepsilon_{t+1}. \quad (3.1)$$

HARQ (Bollerslev et al., 2016) adds the RQ interaction to make the daily coefficient state-dependent:

$$\log \text{RV}_{t+1} = \beta_0 + (\beta_d + \beta_{dQ} \sqrt{\text{RQ}_t}) \log \text{RV}_t^{(d)} + \beta_w \log \text{RV}_t^{(w)} + \beta_m \log \text{RV}_t^{(m)} + \varepsilon_{t+1}. \quad (3.2)$$

The estimated  $\hat{\beta}_{dQ}$  is consistently negative: on noisy days (RQ high), the effective daily coefficient  $\beta_d + \beta_{dQ} \sqrt{\text{RQ}_t}$  shrinks toward zero, shifting forecast weight to the weekly and monthly averages.

### 3.2 The HARQ Shrinkage Mechanism

Figure 3.1 illustrates the core idea: RQ acts as a noise gate on the daily signal.

HARQ with five features consistently beats ML models that use dozens of features without noise-awareness (Bollerslev et al., 2016). The mechanism is simple: when the daily RV estimate is unreliable, the model automatically falls back to longer-horizon averages that average out the noise. This is the single most robust finding in the HAR literature, replicated across asset classes and sample periods.

Clean Day (low RQ)

Noisy Day (high RQ)

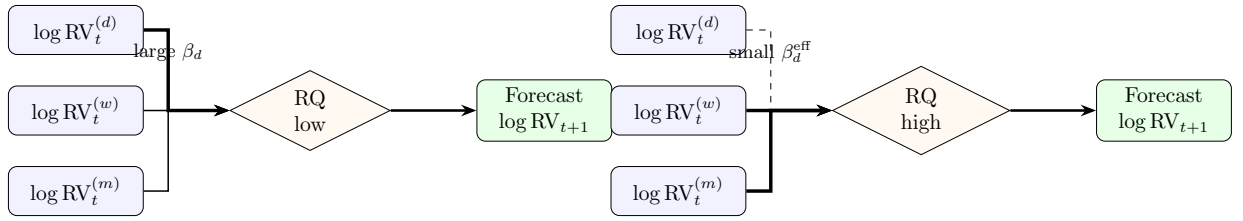


Figure 3.1: HARQ shrinkage mechanism. On clean days the daily reading dominates (thick arrow, left). On noisy days the RQ interaction shrinks the daily coefficient, shifting weight to weekly and monthly averages (thick arrows, right).

### 3.3 Baseline Performance

These five features alone explain 40–60% of next-day log-RV variation across typical equity and index series. RQ requires tick-level returns; we have them for all 34 symbols in the universe (Table 1.1). This is the non-negotiable core of the feature pipeline. Everything in Chapters 4–7 provides marginal improvement on top of this foundation.

Table 3.2: Expected QLIKE improvement from cumulative feature additions (indicative, based on literature benchmarks).

| Layer | Feature Group           | Cumulative QLIKE Gain |
|-------|-------------------------|-----------------------|
| 0     | HAR (3 horizons)        | Baseline              |
| 0     | + RQ interaction (HARQ) | 5–15%                 |
| 1     | + Signed RV, jumps      | 8–20%                 |
| 2     | + VRP, IV skew          | 12–25%                |
| 3     | + Microstructure (OBI)  | 15–30%                |

#### Work in Log-RV Space

Raw RV is right-skewed with heavy tails; log-RV is approximately Gaussian (Andersen et al., 2001). This affects loss functions, residual diagnostics, and model comparison. All features in this guide use log-RV unless stated otherwise. When computing QLIKE, convert back to levels:  $\widehat{RV}_{t+1} = \exp(\widehat{\log RV}_{t+1} + \hat{\sigma}^2/2)$ , where  $\hat{\sigma}^2$  is the residual variance (bias correction for the log-normal retransformation).

#### RQ Interaction: The Single Most Important HAR Extension

Adding the RQ interaction term to baseline HAR yields 5–15% QLIKE improvement from a single feature (Bollerslev et al., 2016). It is the highest-impact extension before adding any external data. Any ML model that does not account for measurement noise in daily RV leaves this gain on the table.

## Chapter 4

# Asymmetric Volatility

Layer 1 decomposes realized variance into signed components and jump variation. Six features, all computed from tick-level 5-minute returns.

### 4.1 Features

Table 4.1: Layer 1 feature set: asymmetric volatility and jump decomposition.

| Feature                      | What It Is                                          | What It Does                                                                                                       | Horizon |
|------------------------------|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|---------|
| $RS_t^-$ (daily)             | $\sum_{i=1}^M r_{t,i}^2 \mathbf{1}(r_{t,i} < 0)$    | Carries $\sim 2\times$ predictive weight of $RS^+$ ; dominant asymmetric signal (Patton and Sheppard, 2015)        | 1d      |
| $RS_t^+$ (daily)             | $\sum_{i=1}^M r_{t,i}^2 \mathbf{1}(r_{t,i} \geq 0)$ | Weaker predictor; provides the contrast that identifies asymmetry                                                  | 1d      |
| $RS_t^{-(w)}$ (weekly)       | $\frac{1}{5} \sum_{j=0}^4 RS_{t-j}^-$               | Persistent downside memory; smooths daily noise in the negative semivariance                                       | 1d–5d   |
| $J_t^-$ (signed neg. jump)   | Large negative moves beyond threshold               | 1–3% QLIKE gain beyond unsigned jumps (Andersen et al., 2007)                                                      | 1d–5d   |
| $C_t$ (continuous variation) | $\max(\text{BPV}_t, 0)$                             | Highly persistent (ACF $\sim 0.6$ – $0.7$ ); workhorse of the decomposition (Barndorff-Nielsen and Shephard, 2004) | All     |
| $J_t$ (jump variation)       | $\max(\text{RV}_t - \text{BPV}_t, 0)$               | Nearly unpredictable (ACF $\sim 0.0$ – $0.1$ ); signals regime breaks                                              | Event   |

### 4.2 The Leverage Effect

Negative returns increase future volatility more than positive returns of equal magnitude. The effect is strongest for equity indices and E-mini futures, where  $RS^-$  carries roughly twice the predictive weight of  $RS^+$  in HAR-type regressions (Patton and Sheppard, 2015). The SHAR model exploits this directly by replacing RV with separate  $RS^+$  and  $RS^-$  regressors at each horizon.

### 4.3 Continuous vs. Jump Variation

Continuous variation  $C_t$  is highly persistent and drives the bulk of out-of-sample forecast accuracy; it behaves like a smoothed version of RV with jump contamination removed. Jump variation  $J_t$  is nearly unpredictable day-to-day, but large positive values flag regime transitions (macro announcements, flash crashes) where the conditional distribution shifts. In practice,  $C_t$  is the forecasting workhorse and  $J_t$  is the alarm:  $C$  enters the model as a persistent regressor at all horizons, while  $J$  triggers recalibration or regime-switch logic. The HAR-CJ model of [Andersen et al. \(2007\)](#) formalizes this split and shows that separating  $C$  from  $J$  yields 1–3% QLIKE improvement over models that use total RV alone.

### 4.4 Cumulative Performance

Layers 0+1 together (11 features: 5 from HAR/HARQ, 6 from this chapter) achieve approximately 70% of the total attainable QLIKE improvement, per literature benchmarks. All semivariances and jump components are computed from tick-level data at 5-minute frequency. Jump significance is assessed via the Lee-Mykland test, which flags individual intraday returns exceeding a time-varying threshold calibrated to local bipower variation.

#### Signed Semivariances: The Cheapest Upgrade

Replacing RV with  $RS^+$  and  $RS^-$  in a HAR regression (the SHAR model) costs zero additional data and yields 3–8% QLIKE improvement ([Patton and Sheppard, 2015](#)). This is the single cheapest feature upgrade after the RQ interaction from Layer 0.

#### Jumps Are Not Predictors

$J_t$  has near-zero autocorrelation. Do not include lagged jump variation as a standard regressor expecting it to forecast future RV. Its value is as a conditioning variable: when  $J_t$  is large, shrink the daily coefficient or trigger a regime indicator. Treating  $J_t$  as a regular feature wastes a degree of freedom and can degrade forecast accuracy.

## Chapter 5

# Options-Implied Features

Layer 2 adds nine features extracted from the SPX implied volatility surface. These are the first forward-looking inputs in the pipeline: they embed market expectations about future variance that backward-looking RV measures cannot capture.

### 5.1 Features

Table 5.1: Layer 2 feature set: options-implied features from the SPX surface.

| Feature                | What It Is                                                                                                                                        | Horizon Impact                            |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| ATM IV (30-day)        | Average of 50-delta put and call IVs from Marquee ERDVOL surface:<br>$IV_t^{\text{ATM}} = \frac{1}{2}(\sigma_{50\Delta P} + \sigma_{50\Delta C})$ | All horizons (Gu et al., 2020)            |
| VRP                    | $VRP_t = (IV_t^{\text{ATM}})^2 - \mathbb{E}_t[\text{RV}_{t+1:t+22}]$ , estimated using HAR forecast as the RV expectation                         | 1w–1m strongest (Bollerslev et al., 2009) |
| 25-Delta Risk Reversal | $RR_t = \sigma_{25\Delta C} - \sigma_{25\Delta P}$ ; measures skew demand for downside protection                                                 | 1–5d                                      |
| Term Structure Slope   | $TS_t = IV_t^{3m} - IV_t^{1m}$ ; positive in calm markets, inverts pre-crisis                                                                     | 1w–1m                                     |
| Butterfly              | $BF_t = \sigma_{25\Delta P} + \sigma_{25\Delta C} - 2 IV_t^{\text{ATM}}$ ; measures tail thickness beyond skew                                    | Crisis detection                          |
| VVIX                   | $VVIX_t$ : implied volatility of VIX options; vol-of-vol signal                                                                                   | 1–5d                                      |
| VIX Term Structure     | $VTSt = F_t^{3m}/VIX_t$ ; ratio $> 1$ = contango (calm), $< 1$ = backwardation (stress)                                                           | Regime signal                             |
| IV–RV Gap              | $Gap_t = IV_t^{\text{ATM}} - \sqrt{\text{RV}_t^{(m)} \times 252}$ ; residual risk premium beyond VRP                                              | 1–5d                                      |
| Event-Implied Vol      | Extracted from the surface around known event dates (FOMC, earnings); isolates the vol the market attributes to the event                         | Pre-event only                            |

### 5.2 Horizon Dependence

Options-implied features contribute almost nothing at  $h = 1$  but become the dominant source of marginal improvement at weekly and monthly horizons. Figure 5.1 summarizes the pattern.

The mechanism is straightforward. At  $h = 1$ , yesterday’s RV is an excellent predictor of tomorrow’s RV because volatility is persistent; options add little incremental information. At  $h = 5$  and  $h = 22$ ,

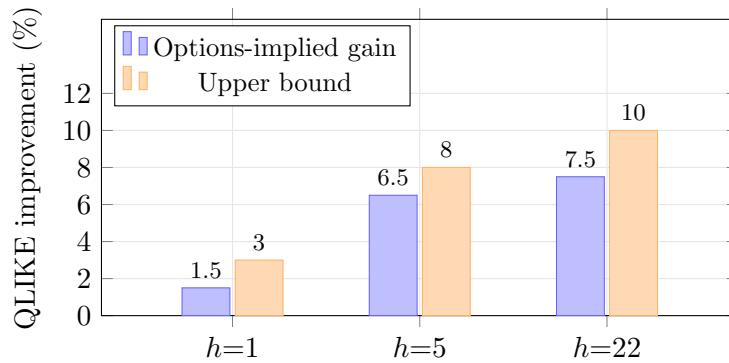


Figure 5.1: QLIKE improvement from adding Layer 2 options features on top of Layers 0–1, by forecast horizon. Ranges:  $h=1$ : 1–3%;  $h=5$ : 5–8%;  $h=22$ : 5–10%. Based on literature benchmarks.

the daily signal decays, and the forward-looking content of options prices fills the gap. Options embed expectations about scheduled events (FOMC, earnings, CPI) that backward-looking RV cannot see.

### 5.3 What We Compute

The full SPX IV surface is sourced from Marquee `ERDVOL_PERCENT_STANDARD`. This provides a complete tenor  $\times$  delta grid from which all nine features are extracted.

The surface covers SPX only. For individual equities, the options-implied features serve as market-wide regime signals (shared across all names in the universe), not stock-specific predictors. This is consistent with the constraint noted in Chapter 2: no single-name IV surfaces are available.

### 5.4 Cumulative Performance

With Layers 0–2 complete (HAR core, asymmetry/jumps, options-implied), the feature set contains approximately 20 features and captures roughly 85% of the forecasting accuracy achievable with the full 120-feature pipeline. The remaining layers (microstructure, cross-asset, engineered interactions) provide diminishing marginal gains.

#### ML Advantage Grows with Horizon

At  $h = 1$ , HAR ties ML on QLIKE. At  $h = 5$  and  $h = 22$ , ML models with options-implied features deliver 10–20% QLIKE improvement over the best linear baseline. The advantage comes from nonlinear interactions between VRP, skew, and term structure that linear models cannot capture (Christensen et al., 2023).

#### VRP Requires a Realized Variance Forecast

The VRP is defined as  $(IV)^2 - \mathbb{E}[RV]$ . The second term is itself a model output (typically HAR). Using VRP as an input feature creates a recursive dependency: the VRP feature quality depends on the baseline forecast quality. Use a simple, fixed HAR estimate for the RV expectation term, not the model being trained.



## Chapter 6

# Microstructure Features

Layer 3 extracts predictive signal from the limit order book and trade flow. These features capture information arrival that daily aggregates cannot see.

### 6.1 Features

Table 6.1: Layer 3 microstructure features.

| Feature                    | What It Is                                                                                 | Evidence                                                    |
|----------------------------|--------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| Price acceleration         | $\sum_i (\Delta \log P_i - \Delta \log P_{i-1})^2$                                         | Single most predictive micro feature (Optiver competition)  |
| WAP log returns            | Returns computed from volume-weighted average price                                        | Less bid-ask bounce than midprice returns                   |
| Order-book imbalance (OBI) | $\frac{\text{bid\_size} - \text{ask\_size}}{\text{bid\_size} + \text{ask\_size}}$ at L1–L5 | (Cartea et al., 2015)                                       |
| Depth ratio                | $\sum \text{bid depths} / \sum \text{ask depths}$                                          | Structural imbalance across book levels                     |
| Market urgency             | $\text{spread} \times \text{OBI}$                                                          | Wide spread + imbalanced book signals imminent move         |
| Spread dynamics            | Level, volatility, and momentum of the bid-ask spread                                      | Increasing spread is predictive of near-term vol            |
| Signed volume flow         | $\text{volume} \times \text{sign}(\text{trade direction})$                                 | Net buying/selling pressure                                 |
| Sub-window RV ratio        | $\text{RV}_{\text{last 5min}} / \text{RV}_{\text{first 5min}}$                             | Within-window acceleration; captures intraday regime shifts |
| VPIN                       | Volume-synced probability of informed trading                                              | (Easley et al., 2012)                                       |

### 6.2 The Optiver Evidence

The Kaggle Optiver Realized Volatility competition (2021) produced the largest public benchmark for microstructure-driven vol forecasting. LightGBM solutions dominated neural networks by roughly 4:1 on the leaderboard. Price acceleration was the single highest-importance feature across top solutions, and sub-window aggregations (e.g., last-5-minute vs. first-5-minute statistics) consistently outperformed whole-window summaries. A representative top-100 solution (91st place) used approximately 600 features, with each base quantity expanded via {level, change, z-score} engineering.

### 6.3 Engineering Principle

For each base quantity in Table 6.1, compute three variants:

Table 6.2: Feature triple expansion.

| Variant | Definition                    | Purpose                    |
|---------|-------------------------------|----------------------------|
| Level   | $x_t$                         | Current state              |
| Change  | $x_t - x_{t-1}$               | Directional momentum       |
| Z-score | $(x_t - \bar{x}_{20})/s_{20}$ | Deviation from recent norm |

This triples the feature count from 9 base quantities to 27. Tree-based models handle the resulting redundancy naturally via split selection; no manual decorrelation is needed.

## 6.4 Data Constraints

L2 order-book depth (five levels of bid/ask size) is available only for the E-mini S&P 500 (ES) via SecDB. Equities receive L1 features only: spread, OBI at best bid/ask, and signed volume flow. Features requiring multi-level depth (depth ratio, market urgency with L2–L5 OBI) are ES-only.

The raw intraday sequences that produce these features also feed the LSTM module (Ch. 10), which consumes them as time-ordered input rather than daily aggregates.

### Lookahead Bias in Microstructure Features

Features for  $RV_{t+1}$  must use only information available at the close of period  $t$ . Timestamp alignment is critical: if  $RV_{t+1}$  covers trading day  $t + 1$ , then features must be computed from data up to and including the close of day  $t$ . A single misaligned sub-window statistic will inflate backtest performance and produce unreproducible results in production.

### Microstructure Features Add Most Value for Trees

Microstructure features are high-cardinality, noisy, and partially redundant. LightGBM exploits them effectively via greedy splits; linear models and shallow networks gain less. The Optiver evidence confirms this: tree ensembles dominated when the feature set was microstructure-heavy.

# Chapter 7

## Cross-Asset Spillovers

Cross-asset features form Layer 4 of the feature pipeline. They capture vol transmission across asset classes, adding 1–5% QLIKE improvement concentrated in regime transitions.

### 7.1 Features

Table 7.1: Layer 4 feature set: cross-asset spillover signals.

| Feature                 | What It Is                                                          | Mechanism                                                                                                      |
|-------------------------|---------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Treasury slope change   | $\Delta(10y - 2y \text{ yield})$                                    | Rate curve inversion precedes equity vol spikes by days                                                        |
| Credit spread momentum  | $\Delta \text{ IG/HY spread or TY futures vol}$                     | Credit stress leads equity vol                                                                                 |
| FX vol (USD/JPY)        | RV of USD/JPY                                                       | Yen carry unwind = global risk-off = equity vol spike                                                          |
| Commodity vol (CL, GC)  | RV of crude oil + gold                                              | Oil = macro uncertainty; gold = flight-to-safety intensity                                                     |
| DY Spillover Index      | VAR(5) variance decomposition on 34-asset RV panel                  | Fraction of vol driven by cross-asset contagion; spikes in crises ( <a href="#">Diebold and Yilmaz, 2012</a> ) |
| Sector-mean RV          | Average RV across same-sector names                                 | Filters idiosyncratic noise                                                                                    |
| VIX-equity corr. regime | Rolling 20-day $\text{corr}(\Delta \text{VIX}, \text{SPX returns})$ | Near $-1$ normally; breaks during regime shifts                                                                |
| Cross-asset RV rank     | Each asset's RV percentile relative to peers                        | Detects outlier dispersion regimes                                                                             |

### 7.2 Impact

Cross-asset features contribute 1–5% QLIKE improvement on average, but the gains concentrate in regime transitions where forecasts are most valuable and hardest to get right. The improvement is small in calm markets and large in the tails: exactly when single-asset models break down, cross-asset signals provide early warning.

### 7.3 Graph-HAR

Graph-HAR extends the standard HAR by adding a neighbor-weighted RV term:

$$\log \text{RV}_{i,t+1} = \beta_0 + \beta_d \log \text{RV}_{i,t}^{(d)} + \beta_w \log \text{RV}_{i,t}^{(w)} + \beta_m \log \text{RV}_{i,t}^{(m)} + \gamma \sum_{k \neq i} W_{ik} \log \text{RV}_{k,t} + \varepsilon_{i,t+1}. \quad (7.1)$$

The term  $\gamma \sum_k W_{ik} \log RV_{k,t}$  captures how asset  $i$ 's vol tomorrow depends partly on its neighbors' vol today. For example, AAPL's forecast improves by incorporating MSFT's realized vol when the two are tightly connected. The weight matrix  $W$  can be correlation-based (simple, static) or learned via a graph neural network (Zhang et al., 2023).

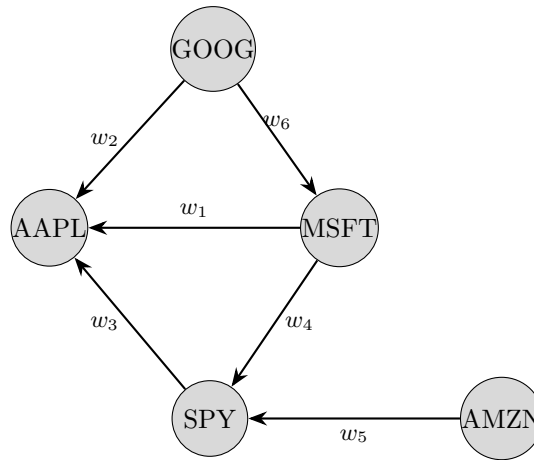


Figure 7.1: Cross-asset vol network. Each node is an asset; directed edges carry weights  $w_k$  determining how much neighbor RV feeds into a given asset's forecast. Weights can be fixed (correlation) or learned (GNN).

#### DY Spillover Index as a Regime Detector

The Diebold and Yilmaz (2012) total spillover index measures the fraction of forecast-error variance attributable to cross-asset shocks. It runs at 30–40% in calm markets and spikes above 70% in crises (2008, 2020). As a feature, its level signals contagion intensity; its rate of change signals regime entry/exit.

#### Nonsynchronous Trading Across Asset Classes

Bonds, FX, and commodities trade on different schedules than equities. Naive daily alignment introduces stale-price bias that inflates apparent lead-lag relationships. Use overlap-window RV (common trading hours only) or the Hayashi-Yoshida estimator for cross-asset covariance.

## 7.4 GS Advantage

Synchronized tick data across asset classes (equities, futures, FX, treasuries) enables intraday lead-lag detection at minute-level granularity. Academic papers rely on daily closes, making sub-day cross-asset transmission invisible and leaving the strongest spillover signals unexploited.

## Chapter 8

# Feature Composition and Selection

This chapter synthesizes Layers 0–4 (Chapters 3–7), adds the remaining layers (calendar, memory, sentiment), and provides the decision framework for which features to use at each forecast horizon.

### 8.1 Calendar, Memory, and Sentiment (Layers 5–7)

Layers 5–7 are individually weak but collectively additive. They are the last 5% of achievable accuracy.

#### Layer 5: Calendar Features

Table 8.1: Calendar features and when they matter.

| Feature                             | When It Matters                                      |
|-------------------------------------|------------------------------------------------------|
| FOMC indicator (−1, 0, +1, +2 days) | Vol compression before announcement, expansion after |
| NFP/CPI indicator                   | Macro release effect on intraday and next-day vol    |
| Options (monthly/quarterly) expiry  | Pinning and gamma unwind                             |
| Quarter-end rebalancing             | Last 3 days of quarter; forced portfolio flows       |
| Earnings proximity                  | Days to next earnings release (single names only)    |
| Time-of-day                         | U-shape: open and close have highest vol             |
| Day-of-week                         | Weakening over time but still detectable             |

#### Layer 6: Memory Features

Table 8.2: Memory features capturing long-range dependence and regime state.

| Feature                     | What It Is                                                                                                    |
|-----------------------------|---------------------------------------------------------------------------------------------------------------|
| Fractionally differenced RV | $(1 - L)^d$ RV with $d \approx 0.35$ – $0.45$ ; preserves long memory while stationary (Lopez de Prado, 2018) |
| Rolling Hurst exponent      | $H < 0.15$ = rough/fast mean-reversion; $H > 0.3$ = trending (Gatheral et al., 2018)                          |
| Vol-of-vol                  | std(RV) over last 22 days; measures instability of the vol process                                            |
| Regime duration             | Days since last $2\sigma$ RV spike; acts as a mean-reversion clock                                            |

#### Layer 7: Sentiment

FinBERT news sentiment and negative news count provide 1–3% QLIKE improvement, concentrated in crisis periods (Audrino et al., 2020). Include these features only if the data pipeline effort is justified

by the target asset class. For index-level forecasts (SPX, SPY), sentiment adds less value than for single names with idiosyncratic news flow.

## 8.2 The Diminishing Returns Curve

Figure 8.1 shows the cumulative accuracy contribution by layer. The first 20 features (Layers 0–2) achieve 85% of attainable accuracy. The remaining 60–100 features add 15%.

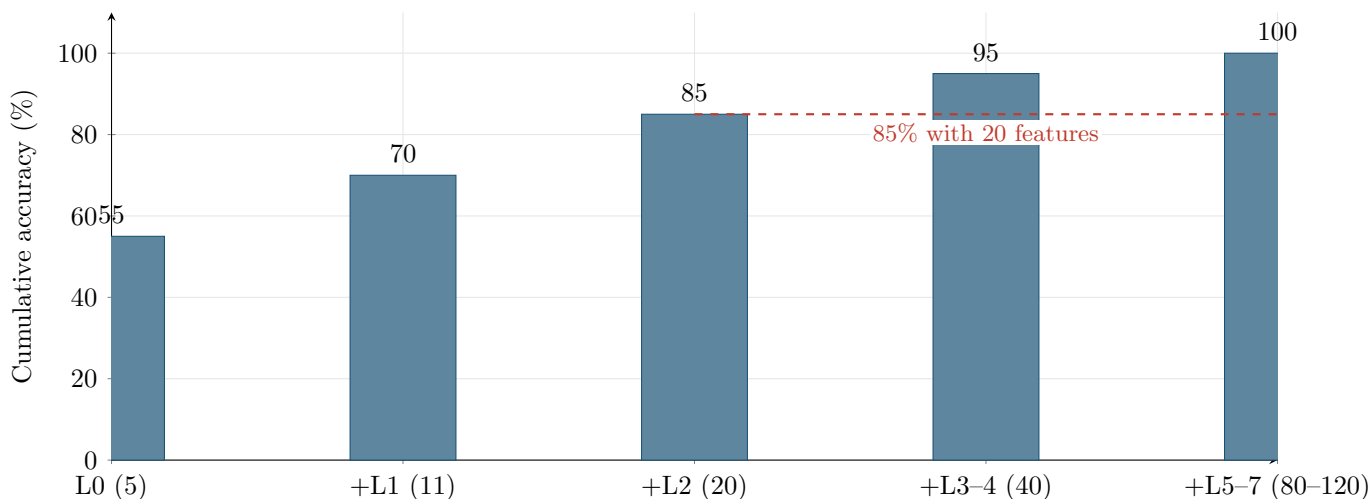


Figure 8.1: Cumulative accuracy by feature layer. Parentheses show total feature count at each stage. The dashed line marks the 85% threshold achievable with Layers 0–2 and a Ridge regression.

## 8.3 Feature Priority by Forecast Horizon

Table 8.3: Dominant features and ML value-add by forecast horizon.

| Horizon              | Dominant Features                                    | Where ML Adds Value                      |
|----------------------|------------------------------------------------------|------------------------------------------|
| Intraday (10min–1hr) | Microstructure (L3): price acceleration, OBI, spread | Trees with 600+ features                 |
| 1 day                | HARQ core (L0) + RQ + asymmetry (L1)                 | HARQ nearly optimal; ML adds ~5%         |
| 1 week               | Options (L2) + cross-asset (L4)                      | VRP + skew: 5–10% over pure RV models    |
| 1 month              | VRP (L2) + macro (L4) + Hurst (L6)                   | Options have max informational advantage |

### The 1-Day Horizon Is the Hardest to Beat

At the 1-day horizon, HARQ with 5 features is nearly optimal. ML models gain ~5% QLIKE improvement at best, and much of that comes from the RQ interaction that HARQ already captures. The largest ML gains are at intraday and weekly+ horizons where linear models lack the capacity to exploit high-dimensional feature sets.

## 8.4 Feature Engineering Principles

- **Triple expansion.** For each base quantity, compute {level, change,  $z$ -score} systematically. This triples the feature count and captures state, direction, and unusualness in a single pass.
- **Horizon-dependent selection.** Drop microstructure features at the monthly horizon (noise at

that scale). Drop calendar features at intraday (FOMC day is already known by 9:30am). Match the feature set to the prediction target.

- **Trees handle redundancy naturally.** Correlated features split across nodes without the multicollinearity problems that plague linear models. No need to pre-decorrelate or drop correlated pairs before training a gradient-boosted ensemble.
- **Feature importance from a single model is unstable.** When features are redundant, a single LightGBM run produces noisy importance rankings that shift across random seeds. Rashomon analysis (Chapter 12) addresses this by examining the set of near-optimal models rather than one point estimate.

## Part III

# Models



## Chapter 9

# LightGBM for Tabular Volatility

LightGBM (Ke et al., 2017) is the primary ML model in the pipeline. It ingests the full engineered feature set from Layers 0–7 and optimizes a custom QLIKE objective.

### 9.1 What Goes In

Table 9.1 summarizes the raw feature counts per layer before expansion. After applying {level, change, z-score} transformations to each base feature, the final input dimensionality is approximately 80–120.

Table 9.1: Feature layers feeding the LightGBM model.

| Layer                                           | Features                                                                           | Count   |
|-------------------------------------------------|------------------------------------------------------------------------------------|---------|
| 0                                               | log RV (d/w/m), RQ, RQ interaction                                                 | 5       |
| 1                                               | $RV^+$ , $RV^-$ , signed jumps, $C$ , $J$                                          | 6       |
| 2                                               | ATM IV, VRP, skew, term structure, VVIX, butterfly, IV–RV gap, stock IV, stock VRP | 9       |
| 3                                               | Price acceleration, OBI, spread, VPIN, Kyle $\lambda$                              | 9       |
| 4                                               | Treasury slope, FX vol, commodity vol, VIX, VIX futures slope, DY spillover        | 6       |
| 5                                               | Calendar: FOMC, OpEx, month/weekday dummies                                        | ~15     |
| 6                                               | Memory: Hurst exponent, fractional $d$ , ACF features                              | ~10     |
| 7                                               | Sentiment: FinBERT scores, attention proxies                                       | ~15–55  |
| <b>Raw total</b>                                |                                                                                    | ~37–57  |
| <b>After {level, change, z-score} expansion</b> |                                                                                    | ~80–120 |

### 9.2 Configuration

Table 9.2 gives the reference configuration, adapted from the Optiver Kaggle competition (91st-percentile solution).

Table 9.2: LightGBM reference configuration.

| Parameter             | Value  | Notes                                |
|-----------------------|--------|--------------------------------------|
| learning_rate         | 0.05   |                                      |
| num_leaves            | 255    |                                      |
| min_data_in_leaf      | 255    | Prevents overfitting on rare regimes |
| n_estimators          | 10,000 | With early stopping                  |
| early_stopping_rounds | 400    |                                      |
| boosting_type         | dart   | Dropout regularization               |
| objective             | Custom | QLIKE loss; not a built-in objective |
| metric                | Custom | QLIKE evaluation for early stopping  |

### 9.3 Custom QLIKE Objective

QLIKE is not a built-in LightGBM loss. Training requires a custom objective function that returns the gradient and Hessian of the QLIKE loss with respect to the predicted value, plus a separate custom evaluation metric for early stopping. QLIKE is robust to noise in the volatility proxy: it ranks forecasts consistently even when RV is measured with error, unlike MSE (Patton, 2011). All training and prediction operate in log-RV space; convert back to levels only for final QLIKE evaluation.

### 9.4 SHAP Interpretability

SHAP values (Lundberg and Lee, 2017) provide feature importance and interaction effects for each prediction. They are required for the GS presentation and for defending model behavior under scrutiny. However, single-model importance measures (gain, permutation, SHAP) are unstable across refits when features are near-substitutes (e.g., VIX, VVIX, ATM IV all proxy for the same latent factor). This instability motivates the Rashomon analysis in Chapter 12, which quantifies importance across the set of near-optimal models rather than relying on a single fit.

#### Baseline First

The fitting scheme for the HAR baseline matters more than the choice of ML model (Bollerslev et al., 2024). A properly fitted HAR using OLS with Newey–West standard errors and an expanding or rolling window is essential before claiming ML improvement. Without this baseline, apparent ML gains may reflect a weak benchmark, not genuine forecasting ability.

#### Why LightGBM

Gradient-boosted trees dominate tabular volatility forecasting benchmarks (Christensen et al., 2023; Branco et al., 2024). They handle mixed feature types (continuous RV lags, categorical calendar dummies, ordinal sentiment scores) without preprocessing, capture nonlinear interactions automatically, and train in seconds on datasets of this size ( $\sim 5,000$  rows  $\times$  120 features).

## Chapter 10

# LSTM for Intraday Sequences

The second model branch operates on sequential intraday data rather than daily tabular features. E-mini S&P 500 Level 2 tick data ( $\sim 4\text{M}$  ticks/day) is compressed into 5-minute (or 1-minute) return bars, each augmented with a limit-order-book snapshot. One training sample is one full trading day's bar sequence. The model's sole task is to produce an independent next-day log RV forecast from this intraday structure.

### 10.1 Architecture

Figure 10.1 shows the data flow from raw ticks to the ensemble blend in Ch. 11.

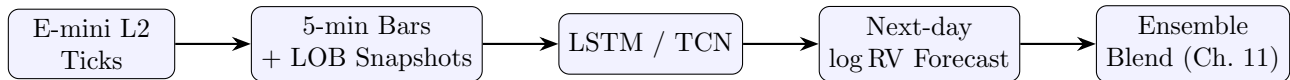


Figure 10.1: Intraday sequence model pipeline.

1. A small LSTM (or TCN) consumes the intraday E-mini sequence for each trading day.
2. Input per time step: 5-min log-return plus LOB snapshot features (bid-ask spread, depth imbalance, trade imbalance, mid-price change).
3. Output: a single scalar, the next-day log RV forecast, produced from the final hidden state via a linear head.
4. This forecast is blended with LightGBM's forecast at the prediction level (Ch. 11).

Table 10.1: Baseline sequence model specifications.

| Component       | LSTM variant              | TCN variant                  |
|-----------------|---------------------------|------------------------------|
| Layers          | 2                         | 4 dilated causal conv blocks |
| Hidden dim      | 64–128                    | 64 channels                  |
| Sequence length | 78 bars (5-min, full day) | 78 bars                      |
| Dropout         | 0.2                       | 0.2 (spatial)                |
| Loss            | QLIKE on log RV           | QLIKE on log RV              |
| Optimizer       | AdamW, cosine LR          | AdamW, cosine LR             |

### 10.2 Embedding Alternative

Instead of producing an independent forecast, the LSTM can export its last-layer hidden state as a fixed-length embedding vector and feed it into LightGBM as additional tabular features (feature-level

stacking). Competition evidence from Optiver (Optiver, 2021) favors prediction-level blending over feature-level stacking, but both approaches should be compared on our data. The embedding path also risks gradient isolation: LightGBM cannot back-propagate into the LSTM, so the embedding is not trained end-to-end for the tabular objective.

### 10.3 Why Deep Learning Here

Four million ticks per day is too rich for hand-engineered aggregations alone to capture fully. Temporal order within the day carries signal: acceleration patterns in returns, depth shifts preceding volatility spikes, and intraday momentum reversals are sequential phenomena that trees cannot exploit from summary statistics. This is the one place in the pipeline where deep learning genuinely adds value over gradient-boosted trees. Optiver competitors worked with 10-minute windows (short sequences where hand-crafted features captured most of the information); our full-day sequences are substantially richer and harder to summarize manually.

#### Diminishing Returns

The LSTM branch will almost certainly contribute less marginal accuracy than the top 20 tabular features in LightGBM. Its value is in capturing residual sequential patterns that tabular aggregations miss, not in replacing the tabular model. Invest proportional effort: get the tabular pipeline right first.

#### DL for Sequences, Trees for Tables

The LSTM operates on sequential intraday data where temporal order matters. LightGBM operates on tabular daily features. Each model gets the data format it handles best. They combine at the prediction level (Ch. 11).

# Chapter 11

## The Ensemble

The pipeline has two model branches: LightGBM (Chapter 9) consuming 80–120 tabular features from Layers 0–7, and an LSTM/TCN (Chapter 10) consuming intraday E-mini L2 sequences. Each branch produces an independent  $\log \text{RV}_{t+h}$  forecast. These forecasts are combined at the prediction level into a single blended output.

### 11.1 Architecture

Figure 11.1 shows the full two-branch architecture converging into the ensemble blend.

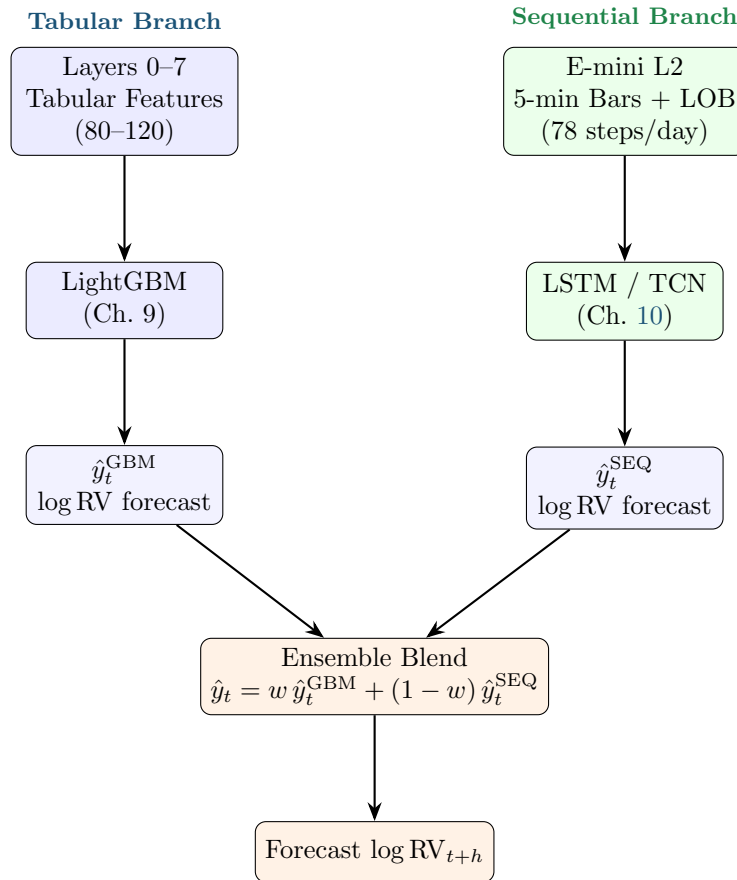


Figure 11.1: Two-branch ensemble architecture. Each branch independently forecasts  $\log \text{RV}_{t+h}$ ; outputs are blended at the prediction level.

## 11.2 Prediction-Level Blending

We blend model *outputs*, not internal representations. Feature-level stacking (feeding LSTM embeddings as LightGBM inputs) breaks gradient isolation and couples model debugging; top-performing Optiver and AmEx Kaggle solutions consistently found that blending independent predictions outperforms feature-level fusion (Optiver, 2021). Blend weights can be static (equal weighting, or a fixed ratio tuned to minimize QLIKE on the validation set) or dynamic (regime-dependent, shifting weight toward the sequential branch during high-microstructure-activity periods). Start with simple validation-tuned static weights; add regime conditioning only if it improves out-of-sample QLIKE.

### Do Not Stack Features

Feeding the LSTM's hidden-state embedding into LightGBM as extra columns is tempting but counterproductive. LightGBM cannot back-propagate into the LSTM, so the embedding is never optimized for the tabular objective. Debugging also becomes harder: a degradation in the blend could originate in either model or in the coupling layer. Keep the branches independent.

## 11.3 Why Two Branches

LightGBM dominates tabular volatility benchmarks (Chapters 3–8 features). LSTM/TCN captures sequential intraday microstructure that tabular summary statistics cannot represent. Giving each model the data format it handles best, then combining, extracts more signal than forcing either model to do both jobs.

### Blend Predictions, Not Features

Each model gets the data format it handles best: trees for tabular, recurrent/convolutional networks for sequences. Combining their independent forecasts at the prediction level outperforms feeding one model's internals into the other.

## Chapter 12

# Interpretable Trees and Rashomon Analysis

LightGBM is a black box. For the GS presentation, we need an interpretable model that can be inspected and defended. Optimal decision trees provide this, and Rashomon analysis reveals which features are genuinely important across the set of near-optimal models.

### 12.1 Optimal Decision Trees

We train a `STreeDPiecewiseLinearRegressor` from `pystreed` on the same Layers 0–7 features used by LightGBM (Chapter 9). Table 12.1 specifies the configuration.

Table 12.1: Optimal decision tree configuration.

| Parameter       | Value                  | Notes                                        |
|-----------------|------------------------|----------------------------------------------|
| Solver          | STreeD                 | DP with elastic-net leaves                   |
| Max depth       | 4–5                    | Tuned via purged blocked $k$ -fold CV        |
| Leaf count      | 8–32                   | Cost-complexity regularization controls this |
| Leaf model      | Elastic net            | Ridge + lasso per leaf partition             |
| Cost-complexity | CV-tuned               | Penalizes $ \text{leaves}(T) $               |
| Features        | Layers 0–7             | Same panel as LightGBM                       |
| Target          | $\log \text{RV}_{t+h}$ | $h \in \{1, 5, 22\}$                         |

Each leaf fits a local elastic-net regression on the features routed to it, so the tree is piecewise-linear rather than piecewise-constant. This gives better accuracy than a constant-leaf tree while remaining fully inspectable: every prediction traces to one root-to-leaf path plus a short linear formula.

Expected accuracy relative to benchmarks:

- $\sim 2\text{--}5\%$  higher MSE than tuned LightGBM (acceptable for interpretability).
- $\sim 10\%$  lower MSE than HAR baseline.
- Comfortably inside the Model Confidence Set when evaluated by QLIKE.

#### Binarization Trap

GOSDT and the original TreeFARMS require binary features. Binarizing continuous vol features (e.g., 50 threshold candidates per feature  $\times$  40 features = 2,000 binary columns) destroys information and inflates runtime exponentially. Use `STreeDPiecewiseLinearRegressor`, which splits on continuous features natively.

## 12.2 Rashomon Analysis Pipeline

The Rashomon set is the collection of all models whose loss is within  $\epsilon$  of the optimum. For decision trees, this set can be enumerated exactly (Xin et al., 2022). Figure 12.1 shows the four-step pipeline.

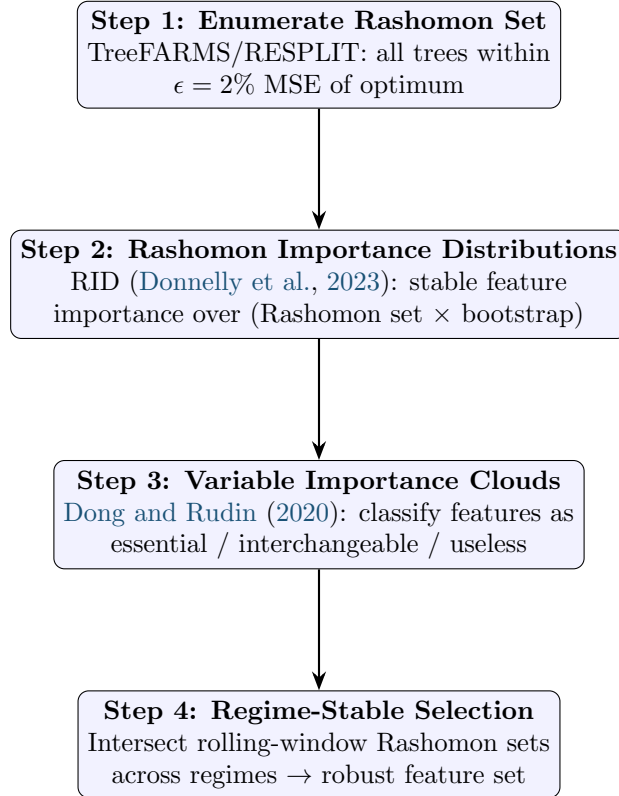


Figure 12.1: Rashomon analysis pipeline. Each step narrows the set of defensible models and the set of robustly important features.

**Step 1.** TreeFARMS (Xin et al., 2022) enumerates every sparse decision tree within  $\epsilon = 2\%$  MSE of the optimal tree. For the vol feature panel ( $\sim 40$  base features,  $\sim 5,000$  rows, depth 4–5), expect  $10^3$ – $10^5$  trees in the Rashomon set. RESPLIT (or SPLIT) provides a faster alternative if enumeration is slow.

**Step 2.** RID (Donnelly et al., 2023) computes a stable importance distribution for each feature across the entire Rashomon set, bootstrapped for confidence intervals. Unlike single-model SHAP or permutation importance, RID has consistency theorems and finite-sample error rates.

**Step 3.** Variable Importance Clouds (Dong and Rudin, 2020) map each feature to its  $[\min, \max]$  importance range across the Rashomon set. Non-overlapping clouds indicate robustly distinct features; overlapping clouds indicate substitutes.

**Step 4.** Train Rashomon sets on rolling windows (e.g., 5-year training, 1-year step). Intersect the sets across windows: features appearing in *every* regime’s Rashomon set are regime-stable; features appearing in only one regime are fragile.

## 12.3 What This Tells Us

- **Genuine importance vs. accidental selection.** Single-model importance (gain, permutation, SHAP) is unstable when features are near-substitutes. RID and VIC identify which features are robustly important across all defensible models.
- **Feature substitution structure.** VIX, VVIX, ATM IV, and the IV–RV spread are near-collinear proxies for the same latent factor. Rashomon analysis reveals which are essential



(appear in every near-optimal tree), which are interchangeable (substitutable without accuracy loss), and which are useless (in no near-optimal tree).

- **Prediction multiplicity.** For any input, the range of forecasts across all Rashomon-set trees quantifies prediction uncertainty due to model choice alone. Report the [5th, 95th] percentile forecast range alongside the point forecast for risk reporting.
- **Post-hoc constraint satisfaction.** Filter the Rashomon set for trees that satisfy additional constraints (monotone in VIX, exclude flagged features,  $\leq 12$  leaves) without retraining. If a compliant tree exists in the set, use it directly.

## 12.4 Evaluation

Table 12.2 defines the evaluation protocol for the interpretable tree and the Rashomon analysis.

Table 12.2: Evaluation protocol for interpretable trees and Rashomon analysis.

| Component          | Protocol                                                                                            |
|--------------------|-----------------------------------------------------------------------------------------------------|
| Accuracy metrics   | Walk-forward MSE, QLIKE, MAE on $\log RV_{t+1}$                                                     |
| Baselines          | HAR, HARQ, LightGBM (Chapter 9)                                                                     |
| Statistical tests  | Diebold–Mariano (Diebold and Mariano, 1995) pair-wise tests; Model Confidence Set                   |
| Acceptable gap     | Optimal tree within 5% MSE of LightGBM; must beat HAR by $\geq 8\%$                                 |
| Regime stress test | Rashomon prediction range during Mar 2020, Volmageddon (Feb 2018); report [5, 95] percentile spread |
| Multiplicity audit | Fraction of test days where Rashomon-set forecast range spans $>1$ QLIKE unit                       |
| Feature stability  | Jaccard similarity of top-10 RID features across rolling windows                                    |

The interpretable tree’s accuracy trade-off is acceptable if: (1) it remains in the Model Confidence Set alongside LightGBM, and (2) the Rashomon prediction range narrows after regime-stable feature selection, confirming that the selected features reduce model ambiguity.

## 12.5 Novelty

No published paper has applied Rashomon methods to any financial time-series problem. The closest applications are cross-sectional credit risk (FICO HELOC dataset) and criminal justice (COMPAS recidivism). Applying Rashomon-set enumeration, RID, and Variable Importance Clouds to realized volatility forecasting is the novel research contribution of this project.

### The Novel Contribution

Rashomon analysis answers: are my features robustly important, or did my model just pick one out of several interchangeable options? This is the novel research contribution.

## Part IV

# Making It Work

# Chapter 13

## Evaluation

This chapter defines how forecasts are scored, how models are compared, and what “better” means quantitatively.

### 13.1 Metrics

Table 13.1 lists every metric used in the project and its role.

| Table 13.1: Forecast evaluation metrics. |                                                           |                              |
|------------------------------------------|-----------------------------------------------------------|------------------------------|
| Metric                                   | What It Measures                                          | Role                         |
| QLIKE                                    | Quasi-likelihood loss; penalizes relative forecast errors | Primary                      |
| MSE                                      | Mean squared error on log-RV                              | Secondary                    |
| MAE                                      | Mean absolute error on log-RV                             | Robustness check             |
| Diebold–Mariano test                     | Statistical significance of pairwise forecast differences | “Is model A better than B?”  |
| Model Confidence Set                     | Set of models not significantly worse than the best       | “Which models are top tier?” |

The primary loss function is QLIKE:

$$\text{QLIKE} = \frac{1}{T} \sum_{t=1}^T \left( \frac{\hat{\sigma}_t^2}{\sigma_t^2} - \log \frac{\hat{\sigma}_t^2}{\sigma_t^2} - 1 \right) \quad (13.1)$$

QLIKE penalizes relative errors, not absolute ones. It ranks forecasters consistently even when RV is measured with noise, because it belongs to the class of loss functions that are robust to imperfect volatility proxies (Patton, 2011).

#### QLIKE Is the Only Loss That Matters for Ranking

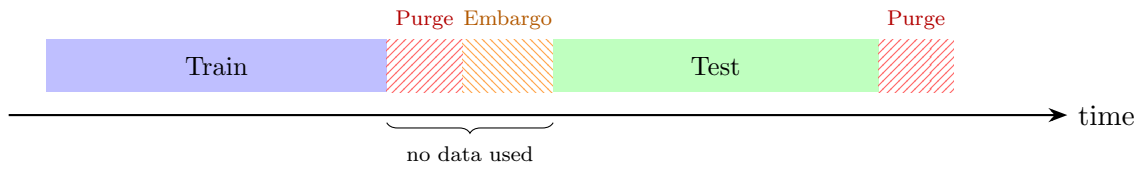
QLIKE is the primary metric for all model selection and comparison decisions. MSE and MAE serve only as secondary diagnostics. The Diebold–Mariano test (Diebold and Mariano, 1995) confirms whether a QLIKE difference is statistically significant; the Model Confidence Set (Hansen et al., 2011) identifies which models belong to the top tier.

### 13.2 Validation Protocol

#### 13.2.1 Purged k-Fold CV with Embargo

Standard cross-validation leaks information in time series because observations near fold boundaries share overlapping feature windows. Purged CV removes observations within the purge window on

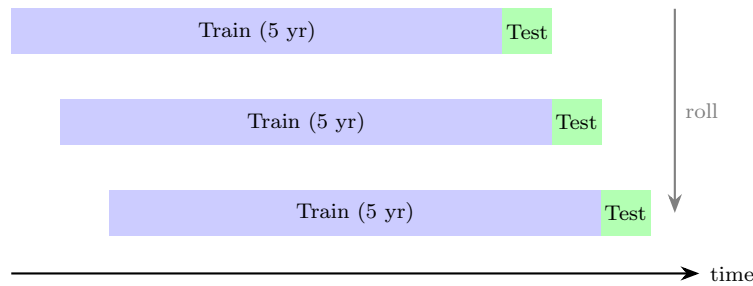
both sides of the train/test boundary. An embargo adds a further gap after each test fold to prevent label leakage from multi-day targets (de Prado, 2018).



We use purged 5-fold CV for hyperparameter tuning. Purge window: equal to the longest feature lookback (22 trading days for the monthly HAR component). Embargo window: equal to the forecast horizon (1 or 5 days).

### 13.2.2 Walk-Forward Evaluation

Walk-forward is the primary out-of-sample evaluation method. Train on a rolling 5-year window, forecast the next period, step forward by one period, and repeat. All reported QLIKE numbers come from this procedure, never from in-sample or CV scores.



## 13.3 Success Target

Three criteria define success:

1. **QLIKE improvement:** 30–80 bps over the HARQ baseline (Bollerslev et al., 2016), averaged across the 35-instrument universe.
2. **Statistical significance:** Diebold–Mariano  $p < 0.05$  vs. each baseline (HAR, HARQ, SHAR, Realized GARCH).
3. **Economic value:** Positive out-of-sample utility gain in a volatility-targeting portfolio (Moreira and Muir, 2017).

#### Train with QLIKE, Not MSE

MSE is dominated by extreme volatility days. A single crisis observation can flip MSE rankings. QLIKE penalizes relative forecast errors and ranks models consistently regardless of the volatility regime (Patton, 2011).

LightGBM does not provide a built-in QLIKE loss. Training requires a custom objective that returns both the gradient and Hessian of the QLIKE loss with respect to the predicted value (see Chapter 9, Section 9.3). If you train with MSE and evaluate with QLIKE, the model optimizes the wrong surface and rankings will not transfer.

## 13.4 Retransformation Bias

Models forecast in log-RV space, but QLIKE and downstream applications require level-space forecasts. Naive exponentiation is biased low by Jensen’s inequality because  $\mathbb{E}[\exp(X)] > \exp(\mathbb{E}[X])$  for any

non-degenerate random variable. The correction is:

$$\widehat{\text{RV}}_{t+1} = \exp\left(\widehat{\log \text{RV}}_{t+1} + \hat{\sigma}_\varepsilon^2/2\right) \quad (13.2)$$

where  $\hat{\sigma}_\varepsilon^2$  is the variance of log-space forecast errors, estimated from a rolling 60-day OOS window.

Table 13.2: Approximate retransformation bias by forecast horizon.

| Horizon  | Bias (without correction) |
|----------|---------------------------|
| $h = 1$  | $\sim 4\%$                |
| $h = 5$  | $\sim 10\%$               |
| $h = 22$ | $\sim 19\%$               |

#### Apply Before Level-Space QLIKE

Apply the correction before computing level-space QLIKE. Without it, every forecast is systematically low, the bias grows with horizon, and the MZ regression (Section 13.5) will show  $a > 0$ .

### 13.5 Mincer–Zarnowitz Regression

QLIKE tells you which model wins; the MZ regression tells you *why* a forecast is bad.

$$\sigma_t^2 = a + b \cdot h_t + \varepsilon_t \quad (13.3)$$

where  $\sigma_t^2$  is realized variance and  $h_t$  is the forecast. An efficient forecast satisfies  $a = 0$ ,  $b = 1$ .

Table 13.3: MZ regression diagnostics.

| Pattern                 | Diagnosis                   | Fix                              |
|-------------------------|-----------------------------|----------------------------------|
| $a > 0$ , $b \approx 1$ | Systematic under-prediction | Check retransformation bias      |
| $a \approx 0$ , $b < 1$ | Forecast too smooth         | Increase reactivity to recent RV |
| $a \approx 0$ , $b > 1$ | Forecast too volatile       | Regularize or increase shrinkage |
| $a = 0$ , $b = 1$       | Efficient forecast          | Report $R^2$                     |

Test the joint hypothesis  $H_0: a = 0$ ,  $b = 1$  with an  $F$ -test.

#### Use HAC Standard Errors

Use Newey–West (HAC) standard errors. Vol forecast errors are serially correlated; OLS standard errors are too small and reject  $H_0$  too often.

### 13.6 Diebold–Mariano Test

A QLIKE improvement means nothing without a  $p$ -value.

Define the loss differential:

$$d_t = L(\sigma_t^2, h_t^A) - L(\sigma_t^2, h_t^B) \quad (13.4)$$

The test statistic is:

$$\text{DM} = \frac{\bar{d}}{\sqrt{\widehat{\text{Var}}(\bar{d})}} \quad (13.5)$$

with Newey–West HAC variance estimator, bandwidth  $\ell = \lfloor T^{1/3} \rfloor$ . Under  $H_0$  of equal predictive ability,  $\text{DM} \xrightarrow{d} \mathcal{N}(0, 1)$ . Reject if  $|\text{DM}| > 1.96$  at 5%.

### Every QLIKE Comparison Needs a DM $p$ -Value

Run pairwise: ML vs HAR, ML vs HARQ, ML vs SHAR, ML vs Realized GARCH. If  $p > 0.05$ , the improvement is not credible.

### Small-Sample Correction

With  $T < 100$ , use the Harvey–Leybourne–Newbold correction (Harvey et al., 1997): replace  $\mathcal{N}(0, 1)$  with  $t_{T-1}$  and apply the finite-sample correction factor.

## 13.7 Model Confidence Set

The DM test compares two models. The MCS compares all of them simultaneously without inflating the false-positive rate.

### Procedure:

1. Start with the full model set  $\mathcal{M}_0$ .
2. Test  $H_0$ : all models in the current set have equal expected loss.
3. If rejected, remove the worst model (highest average loss).
4. Repeat until  $H_0$  is not rejected; survivors form  $\widehat{\mathcal{M}}_\alpha^*$ .

The MCS  $p$ -value for each model is the smallest  $\alpha$  at which the model is excluded. Report it for every model.

Table 13.4: MCS reporting template.

| Model           | QLIKE | DM vs HARQ | MCS $p$ | In MCS <sub>90%</sub> ? |
|-----------------|-------|------------|---------|-------------------------|
| LightGBM (L0-2) | —     | —          | —       | —                       |
| HARQ            | —     | baseline   | 1.000   | Yes                     |
| HAR             | —     | —          | —       | —                       |
| SHAR            | —     | —          | —       | —                       |
| Realized GARCH  | —     | —          | —       | —                       |

### The Credibility Test

The MCS is the credibility test for the presentation. If LightGBM and plain HAR both survive in the 90% MCS, you cannot claim ML superiority. If HAR is excluded and LightGBM survives, that is defensible. Use `arch.bootstrap.MCS` in Python.

## 13.8 Deflated Sharpe Ratio

If the forecast feeds a trading strategy, the backtest Sharpe must survive correction for the number of configurations tested.

Testing  $N$  strategies inflates the expected best Sharpe. Under pure luck,  $\mathbb{E}[\max_i \text{SR}_i] \approx \sqrt{2 \ln N}$ ; for  $N = 20$  this is  $\approx 2.45$ .

$$\text{DSR} = \Phi \left( \frac{(\widehat{\text{SR}} - \text{SR}_0) \sqrt{T-1}}{\sqrt{1 - \hat{\gamma}_3 \widehat{\text{SR}} + \frac{\hat{\gamma}_4 - 1}{4} \widehat{\text{SR}}^2}} \right) \quad (13.6)$$

where  $\widehat{\text{SR}}$  is the observed annualized Sharpe,  $\text{SR}_0 = \sqrt{2 \ln N}$  is the luck threshold,  $T$  is the number of return observations,  $\hat{\gamma}_3$  is skewness, and  $\hat{\gamma}_4$  is kurtosis.

Decision:  $\text{DSR} > 0.95$  is credible.

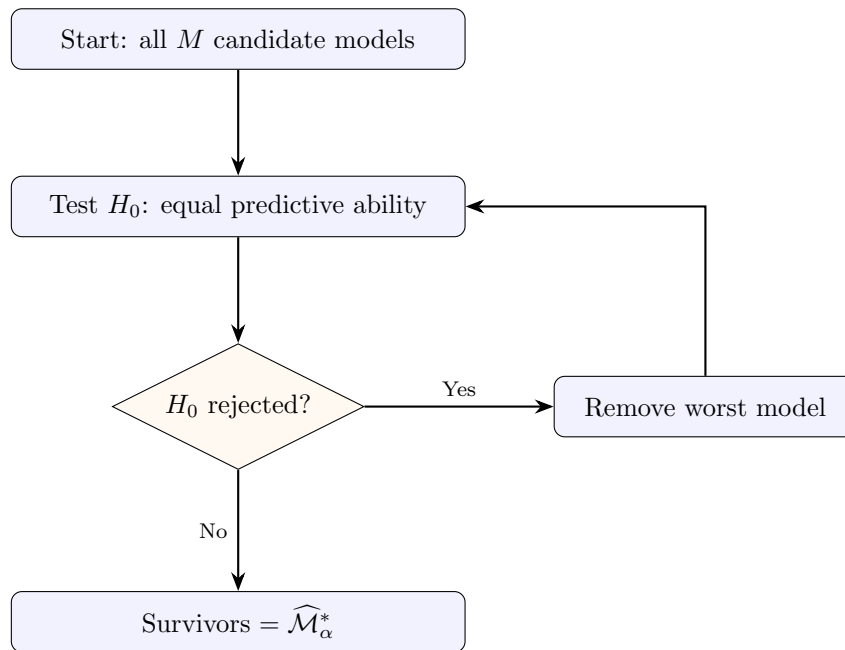


Figure 13.1: MCS elimination procedure.

### Every Experiment Counts

Every experiment counts as a trial. Every hyperparameter grid point, every feature set, every quick look increments  $N$ . Log experiments from day one.

## 13.9 Evaluation Pitfalls

Six errors that invalidate results.

### What Invalidates Your Results

1. **Random  $k$ -fold on time series.** Trains on the future. Always purged CV or walk-forward.
2. **QLIKE improvement without DM test.** A 3% gain with  $p = 0.12$  is noise.
3. **Sharpe without DSR.** A Sharpe of 1.5 from 20 experiments is below the luck threshold.
4. **Training on one regime, testing on another.** 2015–2019 train, 2020 test is a stress test, not general evaluation.
5. **Lookahead in features.** Day- $t$  VIX close for day- $t$  RV is look-ahead. All features must precede the forecast cutoff (Chapter 15).
6. **Tiny improvements without economic significance.** 0.5% QLIKE gain is statistically real but economically meaningless after costs.

## 13.10 Evaluation Workflow

Every forecasting experiment follows this pipeline.

### No Guarantees, Only Evidence

This workflow does not guarantee a good forecast. It guarantees that if you find one, the evidence survives review.

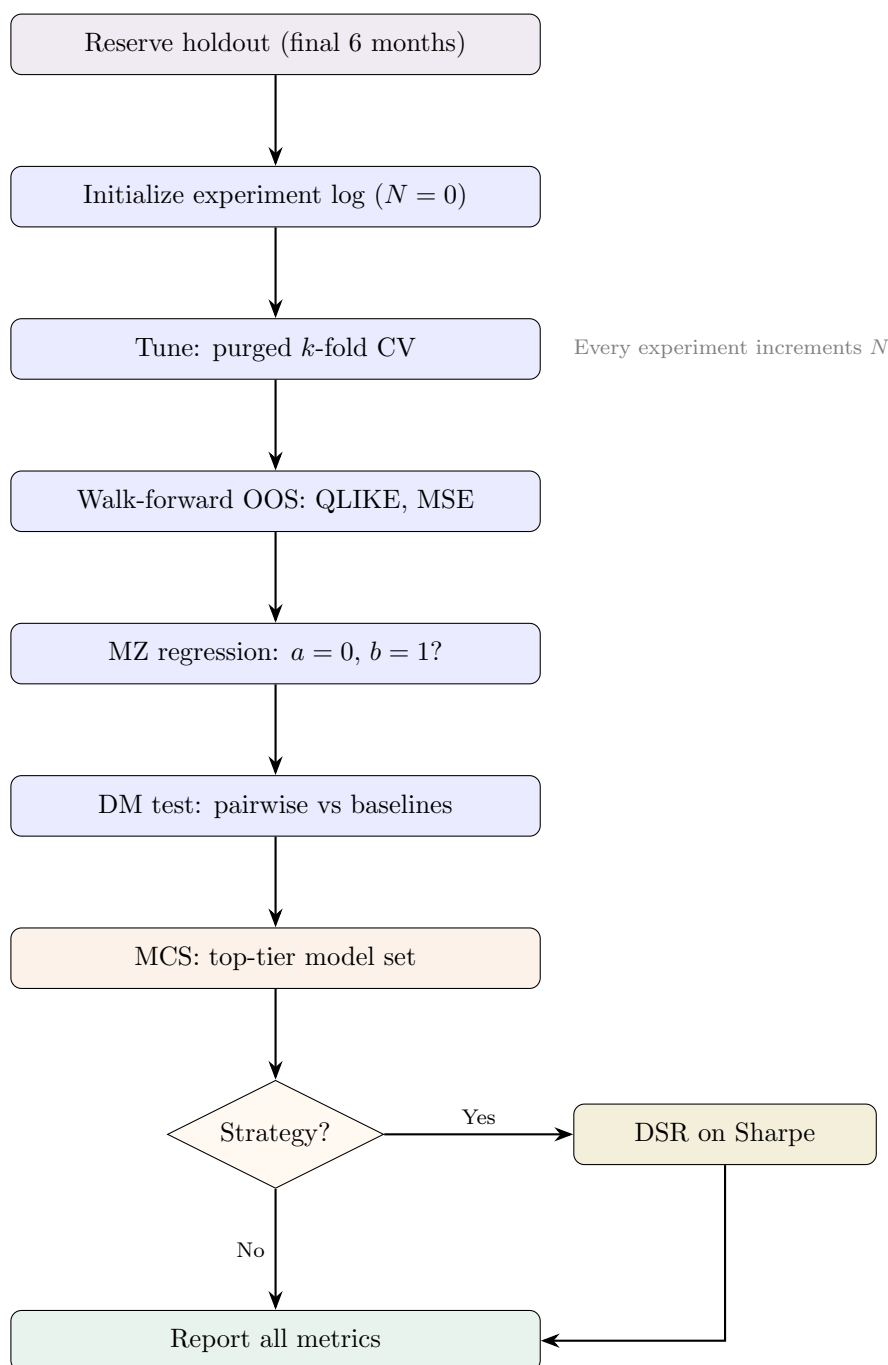


Figure 13.2: Evaluation workflow. Every forecasting experiment follows this pipeline end to end.

Table 13.5: Evaluation toolkit summary.

| Tool                 | Question It Answers                | Reference                   |
|----------------------|------------------------------------|-----------------------------|
| QLIKE                | Which forecast has lower loss?     | Patton (2011)               |
| Retransformation     | Is my level-space forecast biased? | Patton (2011)               |
| Mincer–Zarnowitz     | Is the forecast calibrated?        | Mincer and Zarnowitz (1969) |
| Diebold–Mariano      | Is the improvement significant?    | Diebold and Mariano (1995)  |
| Model Confidence Set | Which models are top tier?         | Hansen et al. (2011)        |
| Purged $k$ -fold CV  | Am I leaking future info?          | de Prado (2018)             |
| Deflated Sharpe      | Is the backtest Sharpe real?       | Bailey and de Prado (2014)  |



# Chapter 14

## The Complete Pipeline

This chapter assembles the data sources (Chapter 2), feature layers (Chapters 3–8), models (Chapters 9–12), and evaluation framework (Chapter 13) into the end-to-end system and defines the implementation roadmap.

### 14.1 End-to-End System Diagram

Figure 14.1 shows every component from raw data to scored forecasts.

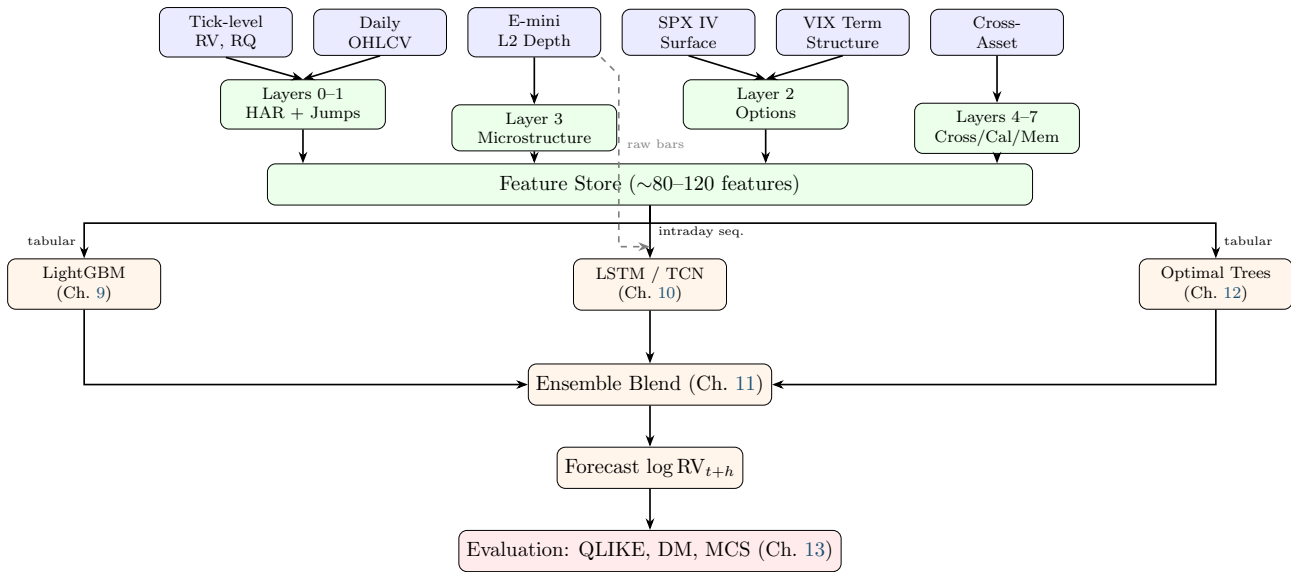


Figure 14.1: End-to-end pipeline from raw data to scored forecasts. Blue: data sources. Green: feature computation and storage. Orange: models and blending. Red: evaluation. Dashed arrow indicates the LSTM also receives raw intraday bar sequences directly.

### 14.2 Implementation Order

Each step produces a standalone, reportable result. No step depends on a later step being complete.

#### Steps 1–2 Are the Critical Path

If Step 1 (HARQ/SHAR baseline) is weak, every subsequent QLIKE comparison is misleading (Bollerslev et al., 2024). Step 2 (adding options features via LightGBM) produces the first genuine ML-vs-baseline comparison. Getting these two steps right matters more than reaching Step 5.

Table 14.1: Implementation roadmap. Each step is independently reportable.

| Step | What                    | Features                 | Model         | Deliverable                          |
|------|-------------------------|--------------------------|---------------|--------------------------------------|
| 1    | HARQ + SHAR baseline    | Layers 0–1 (11 features) | OLS / Ridge   | Walk-forward QLIKE; baseline table   |
| 2    | Add options layer       | Layer 2 (~20 total)      | LightGBM      | QLIKE lift vs. Step 1; SHAP summary  |
| 3    | Cross-asset + spillover | Layer 4 (~30 total)      | LightGBM      | Spillover contribution analysis      |
| 4    | E-mini microstructure   | Layer 3 (separate)       | LSTM / TCN    | Standalone intraday forecast         |
| 5    | Full feature set        | Layers 5–7 (~80–120)     | Ensemble      | Final QLIKE; DM tests; MCS           |
| 6    | Rashomon analysis       | Same as Step 5           | Optimal Trees | Variable importance stability report |

### 14.3 Re-training and Monitoring

Retrain all models weekly on a rolling 5-year window. After each retrain, compute the Rashomon set and record the Jaccard similarity of the top-10 features compared to the previous window; a drop below 0.6 signals feature-importance drift and warrants investigation. Monitor out-of-sample QLIKE on a trailing 20-day basis; alert if it degrades more than 10% relative to the trailing 60-day average. Log every retrain’s feature ranking and QLIKE for post-hoc regime analysis.

### 14.4 Lookahead Bias Checklist

Table 14.2 lists the four primary sources of lookahead bias in volatility forecasting systems.

Table 14.2: Lookahead bias sources and prevention rules.

| Source            | Pitfall                                                           | Rule                                                                      |
|-------------------|-------------------------------------------------------------------|---------------------------------------------------------------------------|
| Realized measures | Intraday returns from the target day leak into features           | Features for $RV_{t+1}$ use only information $\leq t$                     |
| Microstructure    | Full-day LOB features include the close                           | Truncate intraday sequences at $t - \epsilon$ ; align timestamps strictly |
| Options surface   | Intraday surface changes reflect information about the target day | Use end-of-day surface from day $t$ for the day- $t+1$ prediction         |
| Cross-asset       | Mixed frequencies across asset classes                            | Synchronize all inputs to the same end-of-day cutoff                      |

#### Lookahead Bias

The single most common error in financial ML research. Every feature must be computable strictly before the forecast target period begins. When in doubt, add a one-day lag. A lookahead-contaminated model will show excellent in-sample QLIKE that vanishes out of sample, wasting weeks of development time before the bug is identified.

**Part V**

**The Build**

# Chapter 15

## The Data-to-Feature Pipeline

Every feature in the model traces back to one of the six raw data sources in Chapter 2. This chapter maps that lineage: source to measure to feature. For measure formulas, see Chapters 3–7. For selection rationale, see Chapter 8.

### 15.1 Data Lineage

Figure 15.1 shows how data narrows and expands through the pipeline. Six raw sources produce approximately 18 daily scalar measures. These measures pass through lagging, rolling-window aggregation, and triple expansion to produce the final 80–120 feature matrix.

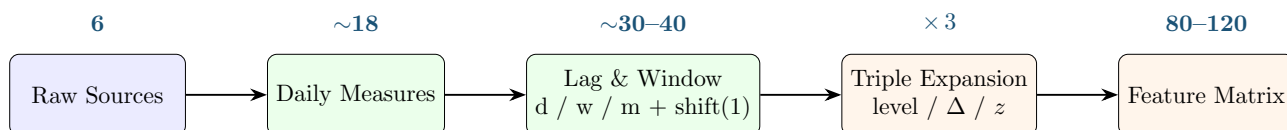


Figure 15.1: Data lineage funnel. Blue: raw inputs (Ch. 2). Green: intermediate measures (Chs. 3–7). Orange: model-ready features. Counts show approximate dimensionality at each stage.

### 15.2 Complete Feature Matrix

Table 15.1 lists every feature in the planned feature set. Read across any row to trace a feature from its source measure through to its final form. Features with d/w/m variants share the same derivation pattern and are collapsed into one row.

Table 15.1: Complete feature matrix with source lineage. Features marked d/w/m apply the same derivation at daily, weekly (5d), and monthly (22d) horizons.

| Layer | Feature                | Source Measure       | Derivation                            | Expansion |
|-------|------------------------|----------------------|---------------------------------------|-----------|
| 0     | log_rv_d/w/m           | rv                   | log [+ rolling mean 5d/22d], shift(1) | —         |
| 0     | sqrt_rq_d              | rq                   | $\sqrt{\cdot}$ , shift(1)             | —         |
| 0     | rq_rv_interaction      | rq, rv               | $\sqrt{rq} \cdot \log(rv)$ , shift(1) | —         |
| 0     | overnight_return       | open, close (TSDB)   | $\log(open_t/close_{t-1})$ , shift(1) | —         |
| 1     | log_rs_positive_d/w/m  | rs_positive          | log [+ rolling mean 5d/22d], shift(1) | —         |
| 1     | log_rs_negative_d/w/m  | rs_negative          | log [+ rolling mean 5d/22d], shift(1) | —         |
| 1     | log_bpv_d/w            | bpv                  | log [+ rolling mean 5d], shift(1)     | —         |
| 1     | log_jump_variation_d   | jump_variation       | log, shift(1)                         | —         |
| 1     | log_continuous_var_d/w | continuous_variation | log [+ rolling mean 5d], shift(1)     | —         |

Continued on next page

Table 15.1 (continued)

| Layer | Feature             | Source Measure      | Derivation                                                                                                | Expansion           |
|-------|---------------------|---------------------|-----------------------------------------------------------------------------------------------------------|---------------------|
| 1     | signed_return_d     | close (TSDB)        | $\log(\text{close}_t/\text{close}_{t-1})$ , shift(1)                                                      | —                   |
| NR    | log_rk_d/w          | rk (tick prices)    | $\log$ [+ rolling mean 5d], shift(1)                                                                      | —                   |
| NR    | noise_gap_d/w       | noise_gap           | [rolling mean 5d], shift(1)                                                                               | —                   |
| 2     | atm_iv_1m, _3m      | atm_iv (Marquee)    | shift(1)                                                                                                  | lev/ $\Delta$ / $z$ |
| 2     | vrp                 | atm_iv, rv          | $IV^2 - RV$ , shift(1)                                                                                    | lev/ $\Delta$ / $z$ |
| 2     | skew_1m             | skew (Marquee)      | shift(1)                                                                                                  | lev/ $\Delta$ / $z$ |
| 2     | term_slope          | atm_iv (3m, 1m)     | $ATM_{3m} - ATM_{1m}$ , shift(1)                                                                          | lev/ $\Delta$ / $z$ |
| 2     | butterfly_1m        | skew, atm_iv        | $0.5(IV_{25\delta P} + IV_{25\delta C}) - IV_{ATM}$ , shift(1)                                            | lev/ $\Delta$ / $z$ |
| 2     | vvix                | VVIX (TSDB)         | shift(1)                                                                                                  | lev/ $\Delta$ / $z$ |
| 2     | iv_rv_gap           | atm_iv, rv          | $IV - \sqrt{RV \times 252}$ , shift(1)                                                                    | lev/ $\Delta$ / $z$ |
| 2     | stock_atm_iv        | EDRVOL (Marquee)    | shift(1)                                                                                                  | lev/ $\Delta$ / $z$ |
| 2     | stock_vrp           | stock_atm_iv, rv    | stock $IV^2 - RV$ , shift(1)                                                                              | lev/ $\Delta$ / $z$ |
| 3     | price_acceleration  | E-mini mid-price    | 2nd derivative (win=50), daily agg, shift(1)                                                              | lev/ $\Delta$ / $z$ |
| 3     | obi                 | E-mini L2 bid/ask   | $(\Sigma \text{bid} - \Sigma \text{ask}) / (\Sigma \text{bid} + \Sigma \text{ask})$ , daily agg, shift(1) | lev/ $\Delta$ / $z$ |
| 3     | depth_ratio         | E-mini L2 depth     | $\log(\text{bid depth}/\text{ask depth})$ , daily agg, shift(1)                                           | lev/ $\Delta$ / $z$ |
| 3     | spread_mean/std     | E-mini bid/ask      | mean/std spread (bps), shift(1)                                                                           | lev/ $\Delta$ / $z$ |
| 3     | vpin                | E-mini trades       | VPIN algorithm, shift(1)                                                                                  | lev/ $\Delta$ / $z$ |
| 3     | kyle_lambda         | E-mini trades       | regress( $\Delta \text{mid}$ , signed vol), shift(1)                                                      | lev/ $\Delta$ / $z$ |
| 4     | treasury_slope      | 10y, 2y note prices | 10y - 2y price proxy, shift(1)                                                                            | lev/ $\Delta$ / $z$ |
| 4     | fx_vol              | USD/JPY, EUR/USD    | annualized rolling RV (22d), shift(1)                                                                     | lev/ $\Delta$ / $z$ |
| 4     | commodity_vol       | CL, GC (TSDB)       | annualized rolling RV (22d), shift(1)                                                                     | lev/ $\Delta$ / $z$ |
| 4     | vix_level           | VIX close (TSDB)    | shift(1)                                                                                                  | lev/ $\Delta$ / $z$ |
| 4     | vix_futures_slope   | VX1, VX2 (TSDB)     | $VX2 - VX1$ , shift(1)                                                                                    | lev/ $\Delta$ / $z$ |
| 4     | dy_spillover        | panel of RVs (34)   | DY FEVD ( $h=10, p=4$ ), shift(1)                                                                         | lev/ $\Delta$ / $z$ |
| 5     | fomc_proximity      | FOMC calendar       | days to next FOMC, shift(1)                                                                               | —                   |
| 5     | nfp_proximity       | NFP calendar        | days to next NFP, shift(1)                                                                                | —                   |
| 5     | opex_proximity      | calendar math       | days to next monthly OpEx, shift(1)                                                                       | —                   |
| 5     | earnings_proximity  | earnings calendar   | days to next earnings, shift(1)                                                                           | —                   |
| 5     | day_of_week         | date                | categorical encoding                                                                                      | —                   |
| 5     | month               | date                | categorical encoding                                                                                      | —                   |
| 6     | frac_diff_rv        | rv                  | $(1 - L)^d$ , $d \approx 0.35-0.45$ , shift(1)                                                            | lev/ $\Delta$ / $z$ |
| 6     | hurst_exponent      | rv                  | rolling Hurst (22d), shift(1)                                                                             | lev/ $\Delta$ / $z$ |
| 6     | vol_of_vol          | rv                  | std(RV) over 22d, shift(1)                                                                                | lev/ $\Delta$ / $z$ |
| 6     | regime_duration     | rv                  | days since last $2\sigma$ spike, shift(1)                                                                 | —                   |
| 7     | finbert_sentiment   | news text           | daily FinBERT score, shift(1)                                                                             | lev/ $\Delta$ / $z$ |
| 7     | negative_news_count | news text           | count of negative articles, shift(1)                                                                      | —                   |

**Notes.** “NR” = noise-robust estimators computed from tick-level log prices, not 5-min bars. “Expansion” shows which features receive the {level, change,  $z$ -score} triple expansion for LightGBM (Chapter 8). Features marked “—” are used as-is. Rolling means compute the average in variance space first, then take log (Corsi, 2009). **noise\_gap** is a ratio, not log-transformed. Treasury slope uses note prices (US10YT=RR, US2YT=RR) as a proxy; true yield tickers are not available in TSDB. EUR/USD is sourced from Marquee FXIVOL, not TSDB. CL and GC require manual front-month contract roll (e.g. CLM26, GCM26); generic continuous contracts are not in TSDB. Deferred features (event-implied vol, sector-mean RV, cross-asset RV rank, WAP log returns, signed volume flow) can be added in later iterations. Every derivation must include shift(1) or equivalent; any feature whose derivation omits an explicit lag uses information from the forecast target period. The most subtle violations come from rolling windows that include day  $t$  when predicting day  $t+1$ . For layer-level summaries and feature counts per model family, see Chapter 9 Table 9.1.

**Every Row Traces Back to Source**

The feature matrix is the blueprint. To understand any feature, read across: the source measure tells you where the data comes from (Chapter 2), the derivation tells you the transformation chain, and the expansion column tells you what LightGBM sees.

# Chapter 16

## System Architecture

Chapter 11 presents prediction blending: two independent model branches combined at the forecast level. This chapter presents two alternative ensemble architectures, feature stacking and residual stacking, and compares all three.

### 16.1 Three Architectures

Figure 16.1 shows the data and prediction flow for each architecture side by side.

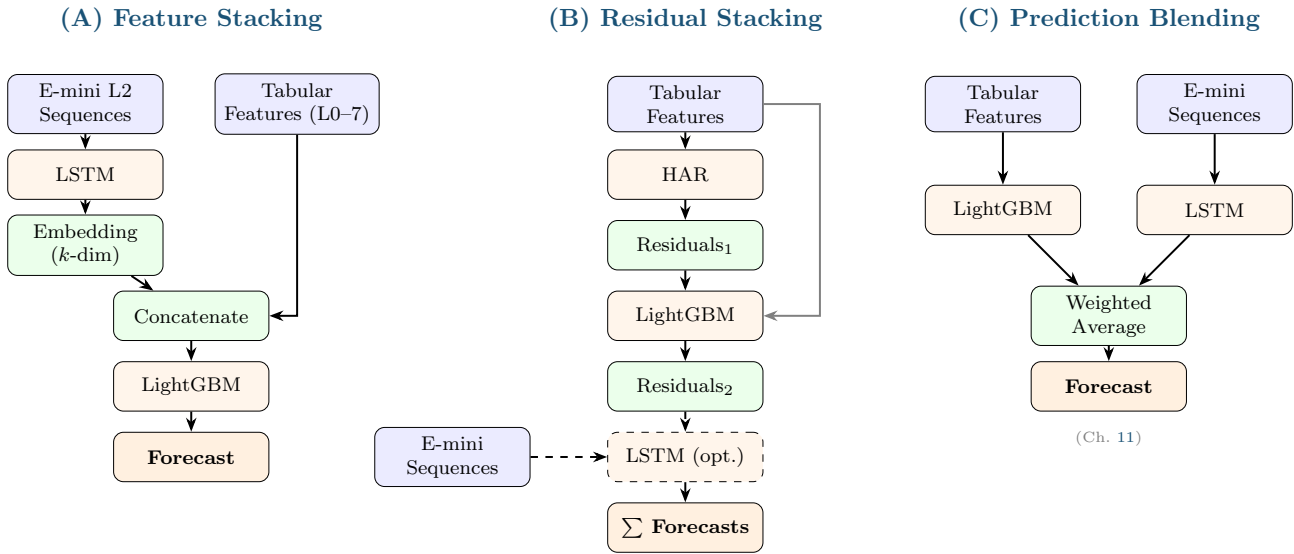


Figure 16.1: Three ensemble architectures. (A) LSTM embedding concatenated with tabular features for a single LightGBM. (B) Each model trains on residuals from the prior stage; final forecast is the sum. (C) Independent models blended at prediction level (Chapter 11).

### 16.2 Feature Stacking

The LSTM processes E-mini L2 5-min bars and LOB features (78 time steps per day) and produces a  $k$ -dimensional embedding vector (default  $k = 32$ , or  $k = 1$  for a scalar forecast). This embedding is concatenated with the  $\sim 80$ – $120$  tabular features from Layers 0–7 to form a single expanded feature set. LightGBM then trains on the combined input.

The approach has one fundamental problem: LightGBM cannot back-propagate gradients through the LSTM, so the embedding is never optimized for the tree’s QLIKE objective. The LSTM learns representations that minimize its own loss, which may not be what the tree needs.

Table 16.1: Feature stacking: pros and cons.

|             |                                                                                                                                                                            |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Pros</b> | Single training pass; LSTM learns representations the tree can exploit                                                                                                     |
| <b>Cons</b> | Gradient isolation (tree cannot backprop into LSTM); embedding not optimized for tree objective; debugging harder; no RV literature demonstrates this beating alternatives |

### 16.3 Residual Stacking

Stage 1: HAR baseline (OLS) produces a forecast and residuals. Stage 2: LightGBM trains on Stage 1 residuals with the full tabular feature set. Stage 3 (optional): LSTM trains on Stage 2 residuals from E-mini sequences. The final forecast is the sum of all stage forecasts.

Each model specializes by construction. HAR captures the multi-scale autoregressive structure that dominates at all horizons. LightGBM captures nonlinear patterns the HAR misses (regime interactions, jump-asymmetry effects). The LSTM, if used, captures whatever sequential dynamics remain in the residuals.

Table 16.2: Residual stacking: pros and cons.

|             |                                                                                                                                                |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Pros</b> | Each model has a distinct role; no gradient isolation; clean residual targets; aligns with HARQ-X direction; supported by recent RV literature |
| <b>Cons</b> | Sequential training (each stage depends on prior); residual signal may be weak at later stages                                                 |

### 16.4 Comparison

Table 16.3: Three-way architecture comparison.

| Dimension        | Feature Stacking                        | Residual Stacking                 | Pred.<br>(Ch. 11)                   | Blending |
|------------------|-----------------------------------------|-----------------------------------|-------------------------------------|----------|
| Complexity       | High (joint training)                   | Moderate (sequential)             | Low (independent)                   |          |
| Gradient flow    | Broken (tree cannot backprop into LSTM) | Clean (residual targets)          | N/A (independent)                   |          |
| Literature       | Weak (no RV paper)                      | Strong (HARQ-X, recent lit.)      | Strong (Optiver, 2021; Bucci, 2020) |          |
| Fallback         | Must retrain tree without embedding     | Drop Stage 3; keep HAR + LightGBM | Drop one model; keep the other      |          |
| Interpretability | Opaque (embedding)                      | Clear (stage contributions)       | Clear (individual forecasts)        |          |

#### Residual Stacking Gives Each Model a Distinct Role

HAR captures multi-scale RV persistence. LightGBM captures nonlinear patterns the HAR misses. LSTM (if used) captures whatever regime dynamics remain. Each model trains on residuals from the prior stage, so roles are distinct by construction.



# Chapter 17

## Modular Pipeline Design

Chapter 14 gives the logical pipeline order. Chapter 16 describes the ensemble model architecture. This chapter describes the software design: config-driven, registry-based, every experiment defined by one YAML file.

### 17.1 Pipeline Architecture

Figure 17.1 shows the four pipeline stages and their plug points. TRAIN is the only modular stage; features, models, and CV strategies resolve from registries at runtime.

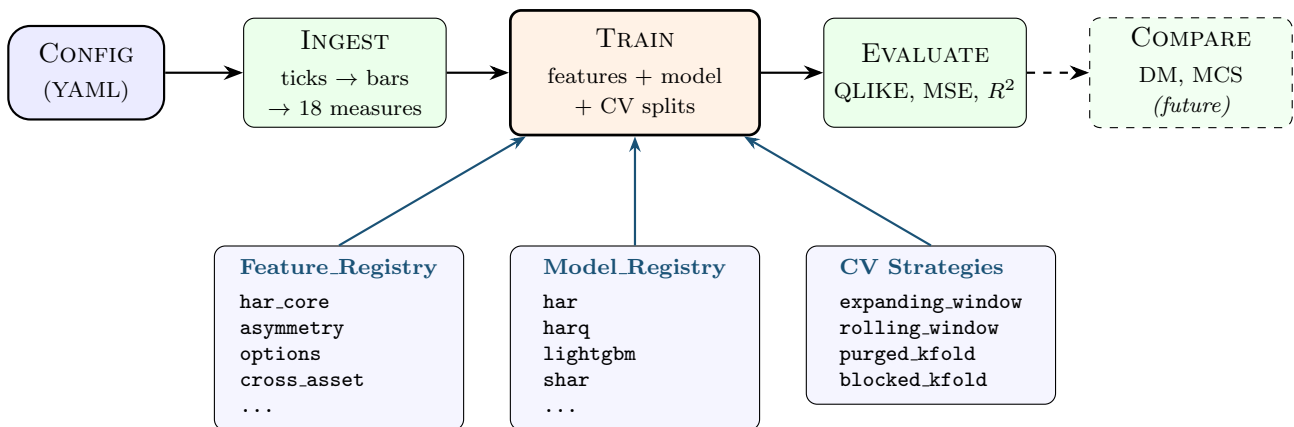


Figure 17.1: Pipeline architecture with plug points. CONFIG parameterises all stages. TRAIN resolves features, models, and CV strategies from registries at runtime. COMPARE (dashed) is not yet implemented.

### 17.2 Plug Points

#### 17.2.1 Feature Layers

The `feature_layers` list in the config names the layers to compose. The pipeline iterates the list, resolves each name from `FEATURE_REGISTRY`, and calls `.compute()`. See Table 15.1 for the full feature inventory.

#### 17.2.2 Models

The `model.name` field selects the model class from `MODEL_REGISTRY`. All models satisfy the same `VolModel` protocol: `.fit(X, y)` and `.predict(X)`. Model-specific hyperparameters go in `model.params`. See Chapter 9 for LightGBM configuration.

### 17.2.3 Cross-Validation

The `cv.method` field selects the splitter: expanding window, rolling window, purged k-fold, or blocked k-fold. Purge gap is set via `cv.purge_gap`. See Chapter 13 for evaluation methodology.

## 17.3 Configuration

Figure 17.2 shows two experiment configs side by side: a HAR baseline and a full LightGBM run.

| HAR baseline                                                                                                               | LightGBM with options features                                                                                                                                                  |
|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>name: baseline_har</code>                                                                                            | <code>name: lgbm_full</code>                                                                                                                                                    |
| <code>universe: [SPY]</code>                                                                                               | <code>universe: [SPY, AAPL, MSFT, NVDA]</code>                                                                                                                                  |
| <code>horizons: [1, 5, 22]</code>                                                                                          | <code>horizons: [1, 5, 22]</code>                                                                                                                                               |
| <code>feature_layers: [har_core]</code>                                                                                    | <code>feature_layers:</code>                                                                                                                                                    |
| <code>model:</code>                                                                                                        | <ul style="list-style-type: none"> <li><code>- har_core</code></li> <li><code>- asymmetry</code></li> <li><code>- options</code></li> <li><code>- cross_asset</code></li> </ul> |
| <code>cv:</code>                                                                                                           | <code>model:</code>                                                                                                                                                             |
| <ul style="list-style-type: none"> <li><code>method: expanding_window</code></li> <li><code>purge_gap: 5</code></li> </ul> | <ul style="list-style-type: none"> <li><code>name: lightgbm</code></li> <li><code>params: {num_leaves: 31}</code></li> </ul>                                                    |
|                                                                                                                            | <code>cv:</code>                                                                                                                                                                |
|                                                                                                                            | <ul style="list-style-type: none"> <li><code>method: purged_kfold</code></li> <li><code>purge_gap: 22</code></li> </ul>                                                         |

Figure 17.2: Two experiments, same pipeline. Left: HAR baseline with one feature layer. Right: LightGBM with four feature layers and stricter cross-validation.

#### One YAML, One Experiment

The entire experiment (universe, horizons, features, model, CV strategy) is defined by a single config file. No code changes needed to run a different experiment.

#### Registries Require Import

Feature layers and models register via decorators at import time. A new layer or model that is never imported will not appear in the registry. The package `__init__.py` triggers all registration imports.

# Chapter 18

## The Development Plan

Chapter 14 gives the logical order for layering features and models. This chapter gives the actual build order: what to implement first given project priorities and foundation work that must happen before anything else. Priority ordering: trading signal > academic rigor > model novelty.

### 18.1 Milestones

Table 18.1 defines eleven milestones with acceptance criteria and dependencies. M0–M2 cover foundation work already underway; M3–M7 form the minimum viable deliverable; M8–M10 are upside.

### 18.2 Critical Path

Figure 18.1 shows milestone dependencies. The critical path (M3 → M4 → M8 → M10) is highlighted. M5, M6, and M9 branch from M2 and run in parallel with the critical path. M7 requires both M4 (forecasts) and M5 (options features). M10 waits on all prior milestones.

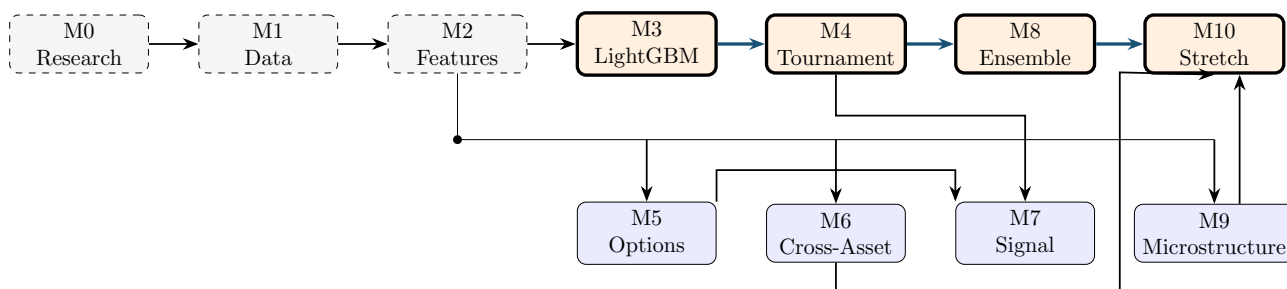


Figure 18.1: Milestone dependencies. Dashed nodes (M0–M2): foundation work. Bold path: critical path (M3–M4–M8–M10). M5, M6, and M9 branch from M2 and run in parallel. M7 requires both M4 (forecasts) and M5 (options features). M10 waits on all prior milestones.

#### M3–M7 Is the Minimum Viable Deliverable

A QLIKE tournament (7 HAR variants + LightGBM, DM tests) plus cross-asset features and a tradeable IV–RV signal with P&L backtest is a presentable result. Everything in M8–M10 is upside. If time runs out after M7, the project has a defensible outcome.

#### M2 Must Be Complete Before M3

The purge gap bug causes silent data leakage for  $h = 22$ . The QLIKE sign convention determines whether the loss function penalizes over-prediction or under-prediction correctly. All results produced before M2 is complete are potentially invalid. Do not skip ahead.

Table 18.1: Development milestones. M3–M7 form the minimum viable deliverable.

| Milestone                            | Acceptance Criteria                                                                                                                                                                                                          | Key Tasks                                                                                                           | Deps   |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|--------|
| <i>Foundation</i>                    |                                                                                                                                                                                                                              |                                                                                                                     |        |
| <b>M0</b> Research & Scoping         | Literature survey (80+ papers); scope defined; feature taxonomy (L0–L5) designed; QLIKE chosen; LightGBM selected; universe (34 symbols) defined                                                                             | Paper review, scope definition, metric/model selection                                                              | —      |
| <b>M1</b> Data Infrastructure        | Tick data via Chunk Store for 34 symbols; daily data from TSDB; IV surface from Marquee; resampling produces 78 bars/day; RV panel outputs 18 measures with parquet caching                                                  | Package scaffold, data integrations, resampling, RV panel builder, config system                                    | M0     |
| <b>M2</b> Feature Engine & Baselines | Layer 0–1 features with no look-ahead; 7 HAR models fit and predict; QLIKE evaluation operational; purge gap $\geq h$ enforced; QLIKE sign matches <a href="#">Patton (2011)</a> ; end-to-end pipeline runs; 390+ tests pass | HAR/asymmetry/noise-robust features, HAR family, metrics, CV splitters, CLI pipeline, purge gap fix, QLIKE sign fix | M1     |
| <i>MVP</i>                           |                                                                                                                                                                                                                              |                                                                                                                     |        |
| <b>M3</b> LightGBM                   | Custom QLIKE objective converges; Optuna finds improved params; walk-forward OOS predictions for 3 horizons                                                                                                                  | QLIKE gradient/hessian, model class, Optuna, walk-forward                                                           | M2     |
| <b>M4</b> Tournament                 | 8 models $\times$ 3 horizons table with DM $p$ -values and Mincer–Zarnowitz efficiency tests on dev universe                                                                                                                 | Run baselines, DM test, MZ regression, tournament table                                                             | M3     |
| <b>M5</b> Layer 2 Options            | Options features produce daily values with no look-ahead; QLIKE lift documented                                                                                                                                              | OptionsLayer, IV surface wiring, validation                                                                         | M2     |
| <b>M6</b> Layers 4–5                 | Cross-asset and calendar features produce daily values; DY spillover index computed; FOMC/NFP/OPEX proximity indicators operational                                                                                          | Treasury slope, FX/commodity vol, DY spillover, event calendars                                                     | M2     |
| <b>M7</b> Signal                     | IV–RV gap signal; equity curve; positive OOS Sharpe                                                                                                                                                                          | Signal logic, P&L back-test, performance metrics                                                                    | M4, M5 |
| <i>Upside</i>                        |                                                                                                                                                                                                                              |                                                                                                                     |        |
| <b>M8</b> Ensemble                   | Residual stacking and prediction blending tested; 10-model tournament table                                                                                                                                                  | Residual stacking, inverse-QLIKE blending                                                                           | M4     |
| <b>M9</b> Microstructure & Sequences | Layer 3 E-mini features (OBI, VPIN, depth); LSTM/TCN on intraday bars; scalar forecast as LightGBM feature evaluated                                                                                                         | Microstructure layer, LSTM architecture, sequence pipeline                                                          | M2     |
| <b>M10</b> Stretch                   | Ordered by impact: regime QLIKE, MCS, Rashomon (TreeFARMS + RID + VIC), STreeD, Layer 6–7 (memory/sentiment), reporting/visualization, full universe, figures                                                                | Each task independent                                                                                               | M4–M9  |

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